

Review

Molecular phenotypes and spatial archetypes: A new framework for cancer-associated fibroblasts

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SUMMARY

Cancer-associated fibroblasts (CAFs) form a dynamic ecosystem that critically influences tumor progression and therapeutic response. Although recent advances in single-cell and spatial omics have uncovered profound stromal diversity, interpreting the mechanistic relevance of this complexity remains a challenge. Here, we propose a more unifying conceptual framework to bridge high-dimensional data with experimental biology. By categorizing CAFs into conserved molecular phenotypes and distinct spatial archetypes, this model illustrates how CAF identities are intimately linked to local tissue contexts. This refined framework brings the complexity of the tumor stroma into greater focus, underscoring the necessary transition from broad stromal targeting toward precision, context-specific modulation. Ultimately, we hope this integrated effort will aid in the collaborative development of next-generation therapies that selectively target pathogenic stroma in cancer to improve patient outcomes.

INTRODUCTION

Fibroblasts were first identified by Rudolf Virchow in 1858 and named by Ernst Ziegler in 1881 to reflect their role in producing connective tissue during wound healing. In healthy organs, they maintain extracellular matrix (ECM) homeostasis, define structural boundaries between parenchyma and stroma, and serve as dynamic first responders that orchestrate wound repair. In malignancy, this homeostatic cycle is subverted. Tumors, long described as “wounds that do not heal”,¹ co-opt conserved reparative programs, resulting in persistent activation of tissue-resident and recruited mesenchymal progenitors collectively referred to as cancer-associated fibroblasts (CAFs). CAFs exhibit functional diversity that evolves throughout tumor pro-

gression.² Within primary tumors and metastases, CAFs orchestrate ECM remodeling; secrete growth factors, cytokines, and extracellular vesicles; modulate metabolism; and establish immunosuppressive niches that facilitate invasion, metastasis, and therapeutic resistance.^{3,4}

Historically, CAF biology was constrained to a limited set of activation markers. This oversimplified classification not only limited our understanding of stromal heterogeneity but also hindered the development of selective therapies. Indeed, early efforts to non-selectively ablate these bulk populations yielded conflicting outcomes, including paradoxical tumor acceleration.^{5,6} This highlighted that canonical broad markers obscure functionally distinct stromal subsets. The advent of single-cell RNA sequencing (scRNA-seq) and spatial transcriptomics

resolved this complexity by revealing that CAFs constitute a landscape of distinct lineages and states, intricately linked to their specific cellular progenitors and spatial niches.^{7–12} Yet high-resolution data have introduced its own challenge: dozens of transcriptional clusters populate the literature across cancer types, and it remains difficult to determine which represent conserved biological entities described under different names versus transient, context-specific adaptations. Resolving these relationships is essential, to clarify whether the observed heterogeneity reflects plastic lineages, fixed populations, or both, and to distinguish CAFs from their progenitors within the tumor microenvironment (TME).

In this review, we aim to integrate the collective observations into a proposed coherent biological model. We focus on conserved molecular phenotypes reported across independent studies: inflammatory CAFs (iCAFs), myofibroblastic CAFs (myCAFs), and antigen-presenting CAFs (apCAFs). By synthesizing lineage-specific evidence with pan-cancer single-cell and spatial atlases, we examine how CAF transcriptional states relate to one another and how their molecular programs are shaped by local niches. We organize this complexity through a hierarchical classification system and a set of recurring spatial archetypes, with the goal of improving reproducibility, identifying conserved therapeutic vulnerabilities, and accelerating the transition of CAF-directed therapies from discovery to the clinic.

ORIGINS OF CAF HETEROGENEITY

Transcriptional and functional heterogeneity among CAFs may be attributed to at least two drivers: context-dependent plasticity and distinct cells of origin. CAFs sense and respond to contextual cues from neighboring cells, producing diverse mediators that can promote or restrain tumor growth.² Within a dynamic TME, positional cues and signaling inputs change over time, generating measurable heterogeneity within this plastic population.

In human breast cancer, analysis of 10 spatially defined cellular niches revealed that FAP⁺ CAF plasticity was reflected by distinct transcriptional signatures potentially shaped by niche cues.⁸ An ADH1B⁺ Detox-iCAF cluster residing in inferred immune-protective niches with FOLR2⁺ macrophages appeared to transition toward an immune-suppressive ECM-myCAF state, close to cancer cells, either directly or indirectly via a Wound-myCAF intermediate under high YAP1-dependent signaling. Trajectory inference and functional assays using primary FAP⁺ CAFs confirmed plasticity between these defined states in several cancer types, and following chemotherapy.^{8,13} In pancreatic ductal adenocarcinoma (PDAC), spatial transcriptomics revealed that myCAFs consistently localize near basal-like cancer cells.¹⁴ Functional assays further demonstrated that pancreas stellate cells (PSCs) adopt a myCAF phenotype mediated by high transforming growth factor β (TGF- β) or an iCAF phenotype mediated by IL-1/JAK-STAT activation.^{12,15,16} Regulatory T cells (Tregs) can serve as key TGF- β sources driving the myCAF state, linking immune contexture to CAF plasticity.¹⁷ Reciprocally, specific myCAF subsets, such as FAP⁺ ANTXR1⁺ myCAFs in breast and ovarian cancer, can directly promote the attraction, survival, and differentiation of Tregs, establishing a mutually reinforcing immunosuppressive loop.^{18,19} In human lung cancer, two CAF populations found at different stages of tumor growth associate

with T cell exclusion via distinct matrix-based mechanisms: alpha-smooth muscle actin (α -SMA)⁺ CAFs with collagen IV fibers are present in early stage tumors, and FAP⁺ α -SMA⁺ CAFs with collagen XI/XII are present in advanced stage tumors.²⁰ This highlights plasticity over time despite conservation of T cell-excluding and Treg-promoting functions. Mouse models show similar dynamic remodeling of CAF transcriptional profiles during lung metastasis progression.²¹ In addition to microenvironmental cues, tumor-intrinsic genetic alterations can also influence CAF composition and state. In breast cancer, *BRCA*-mutated tumors exhibit different CAF compositions compared to *BRCA* wild-type tumors, suggesting an association between cancer genotype and stromal heterogeneity.⁹ Extending this concept mechanistically, loss of *BRCA* function in PDAC cancer cells has been shown to drive a transcriptional shift in PSCs from a myofibroblastic to an immune-regulatory phenotype.²²

Multiple cell types can transition to a CAF phenotype, such that cellular origins intersect with signaling inputs to define CAF fates (Figure 1). CAFs predominantly arise from tissue-resident mesenchymal precursors,²³ though bone marrow-derived precursors have also been identified.²⁴ Among tissue-resident fibroblasts, a universal fibroblast program has been identified across organs and is inferred to be the predominant precursor cells from which CAFs arise in a variety of settings.⁷ These universal fibroblast progenitors, transcriptionally defined by *DPT*, *PI16*, and *COL15A1* at steady state, are thought to transition into CAF states when exposed to tumor-associated fibroinflammatory conditions. Lineage tracing studies using *Dpt* reporter mice confirm tissue-resident universal fibroblasts as major CAF precursors.^{25,26} Additional studies have shown tissue-restricted precursors including Hedgehog-responsive GLI1⁺ fibroblasts in the pancreas²⁷ and stellate cells in the liver and pancreas, give rise to CAFs in both primary tumors and liver metastases.^{28–30} Mesothelial cells lining serous cavities may give rise to major histocompatibility complex class II (MHC-II)-expressing apCAFs in multiple anatomic sites, with functional significance potentially differing across tissues.^{11,31} Some evidence supports a pericyte origin from lineage tracing in squamous carcinoma and lung cancer, as well as computational inference across cancer types,^{32–35} though further tracing is needed to assess pericyte contribution across organs. Endothelial-to-mesenchymal transition has also been proposed but remains to be formally demonstrated in the TME.^{36,37}

Despite this plasticity, recent studies highlight measurable non-redundancy among CAFs of distinct cellular origins. In PDAC, inhibition of certain CAF lineages yields tumors with similar total CAF abundance but measurable phenotypic differences, suggesting numerical but not complete functional compensation.^{25,30} CD105 expression distinguishes two stably separate PDAC CAF populations: CD105[−] CAFs are tumor-suppressive and promote anti-tumor immunity via an adaptive immune-dependent, MHC-II-independent mechanism.³⁸ In contrast, TGF- β -dependent LRRC15⁺ CAFs suppress anti-tumor immunity. While their disruption during early tumor development triggers a compensatory expansion of LRRC15[−] CAFs, targeted ablation of LRRC15⁺ CAFs in established tumors enhances checkpoint blockade efficacy.²⁵ Transcriptional profiling reveals some functional overlap across lineages in matrix production and cytokine expression, but other functions may be lineage-restricted.³⁰ Stellate cells

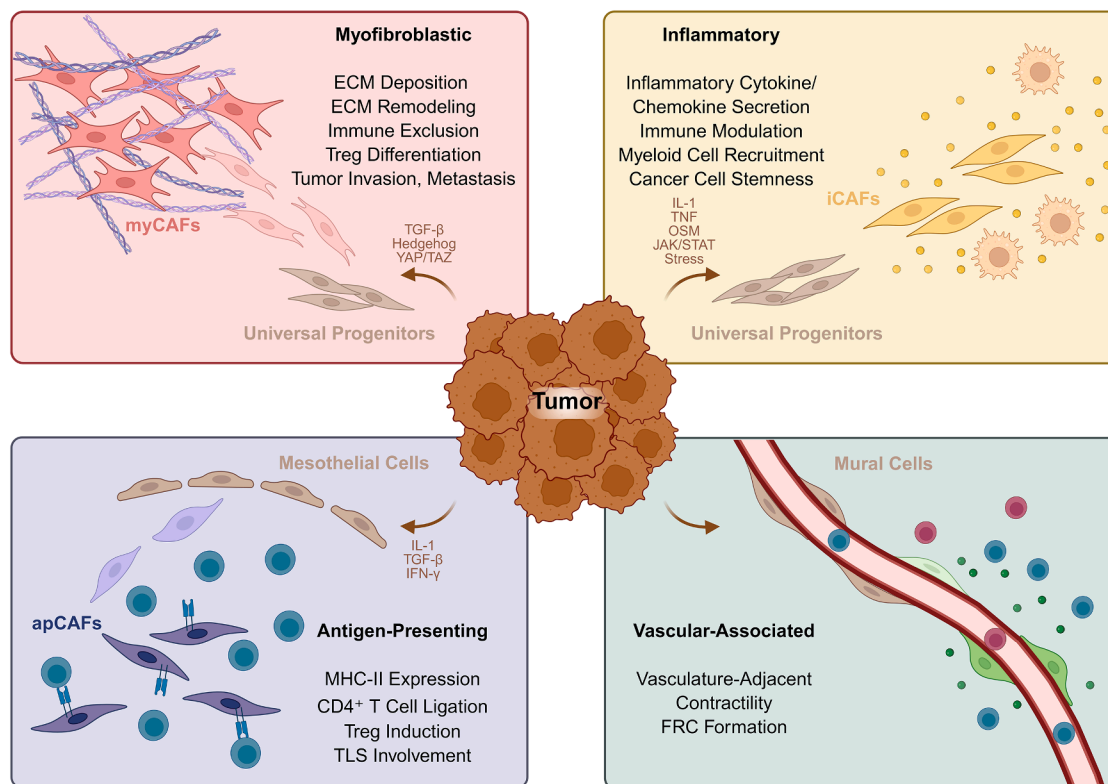


Figure 1. CAF origins and conserved molecular programs

Schematic illustrating how tumor-derived cues drive precursors into four conserved CAF phenotypes. Upper left: universal fibroblast progenitors differentiate into myfibroblastic CAFs (myCAFs) through TGF- β , Hedgehog, and YAP/TAZ signaling. myCAFs are characterized by ECM deposition, ECM remodeling, immune exclusion, Treg differentiation, and promotion of tumor invasion and metastasis. Upper right: universal progenitors alternatively differentiate into inflammatory CAFs (iCAFs) through IL-1, TNF, OSM, JAK/STAT signaling, and stress-response pathways. iCAFs mediate inflammatory cytokine and chemokine secretion, immune modulation, myeloid cell recruitment, and cancer cell stemness. Lower left: mesothelial cells give rise to antigen-presenting CAFs (apCAFs) under IL-1, TGF- β , and IFN- γ stimulation. apCAFs are defined by MHC-II expression, CD4⁺ T cell ligation, Treg induction, and TLS involvement. Lower right: mural cells contribute to vascular-associated CAF programs characterized by vasculature-adjacent positioning, contractility, and FRC formation. The central tumor mass indicates that tumor-derived signals coordinate the activation and functional specialization of CAF phenotypes.

give rise to some but not all myCAFs in PDAC; tumors lacking stellate cell-derived CAFs show similar total collagen but differences in mechanosignaling and specific matrix components such as tenascin C.³⁰ In hepatocellular carcinoma (HCC), stellate cells give rise to both iCAFs and myCAFs depending on local cues, with different genetic perturbations in the same lineage yielding different functional outcomes.²⁹ Together, these findings suggest that cell of origin can constrain the fate potential of CAFs, while some functions may be shared upon sensing of spatial or paracrine cues regardless of origin.

CONSERVED CAF PHENOTYPES

Inflammatory CAFs

iCAFs are defined by a secretory program dominated by inflammatory cytokines and chemokines (e.g., IL-6, LIF, CXCL1, CXCL2, and HGF), alongside low α -SMA and high PDGFR α expression^{16,19,39} (Figure 1). Current models suggest iCAF specification is influenced by IL-1/TGF- β antagonism.¹⁵ Tumor-derived IL-1 α activates NF- κ B and induces LIF, which sustains the iCAF phenotype through a JAK/STAT3 autocrine loop, while TGF- β opposes this by downregulating IL-1R1 via

SMAD2/3, redirecting fibroblasts toward a myCAF fate. This antagonism couples CAF phenotype to spatial positioning, with proximal fibroblasts receiving juxtacrine TGF- β and distal fibroblasts primed by paracrine IL-1.^{8,15,16} Reflecting the likely redundancy of *in vivo* networks, oncostatin M (OSM), tumor necrosis factor (TNF), and IL-17A may synergize with IL-1 to drive iCAF polarization.^{15,40–43} However, the precise *in vivo* compensatory mechanisms among these redundant signals require further investigation.

Stress responses also promote iCAF identity. Hypoxia induces iCAF gene expression via HIF-1 α ,⁴⁴ HSF1 drives iCAFs in BRCA-mutated PDAC,²² glutamine depletion activates MEK-ERK inflammatory signaling,⁴⁵ and NRF2/oxidative stress pathways are enriched in iCAFs across tumor types.³⁵ These findings suggest iCAFs can arise as a general tissue damage response, reinforcing the link between tumors and “wounds that do not heal”. The shared dependence on NF- κ B between iCAF programs and the inflammatory phase of wound repair raises the possibility that tumors co-opt conserved damage-response circuits to buffer against therapeutic insults: iCAF signatures associate with chemoresistance and poor outcomes,^{41,46} CD10⁺GPR77⁺ CAFs sustain cancer stem cell niches via NF- κ B-driven IL-6/IL-8

secretion,⁴⁷ and chemoradiation in rectal cancer triggers IL-1 α -driven iCAF polarization promoting tumor survival.⁴⁶

A remaining challenge is distinguishing bona fide iCAFs from activated fibroblast progenitors. DPT⁺ or PI16⁺ universal progenitors express *CXCL12* and other chemokine genes used in some iCAF definitions, but higher-resolution pan-cancer analyses resolve dedicated iCAF populations with high *IL6*, *LIF*, or *CXCL1* and low progenitor-marker expression.^{7,11,48} Whether these defined phenotypes represent stable populations or transient phases of activation remains unresolved.

Myofibroblastic CAFs

myCAFs are typically the most abundant CAF subtype, often comprising the majority of CAFs in desmoplastic cancers.^{10,11,48} While the specific designation “myCAF” emerged from recent PDAC studies to distinguish them from other populations,¹⁶ this phenotype effectively embodies the classical understanding of CAFs that dominated the field for decades: proximal to cancer cells, contractile, ECM-rich, and α -SMA-high^{10,15,16} (Figure 1). Although TGF- β signaling is one of the primary drivers of myCAF differentiation, additional pathways such as Hedgehog and YAP/TAZ mechanotransduction signaling may contribute to the maintenance of this phenotype.^{15,25,49,50}

The functional contribution of myCAFs to tumor progression remains one of the most intensely debated topics in stromal biology.⁴ Early work suggested α -SMA⁺ stromal cells can restrain tumor progression, but interpretation is complicated by α -SMA expression in mural cells.^{5,51} Recent studies using *Lrat-Cre* to target stellate cells have demonstrated that myCAF-derived ECM is context dependent: in PDAC, *Lrat*⁺ cell-derived collagen restricts tumor growth while the cells themselves promote dissemination,⁵² whereas in HCC, both the *Lrat*⁺ myCAFs and their collagen drive tumor development.²⁹ These conflicting preclinical outcomes not only reflect the context-dependent nature of ECM but also underscore our incomplete understanding of CAF cellular origins. For example, the field may need to resolve the relationship between stellate cells and other progenitors (such as DPT⁺ cells), ensuring that our genetic tools accurately reflect true myCAF biology *in vivo*.

Beyond ECM remodeling, myCAFs promote cancer metastasis through additional mechanisms. EGFR-activated CD90⁺ myCAFs promote PDAC metastasis, potentially by inducing epithelial-mesenchymal transition (EMT) in cancer cells,⁵³ and SPP1-producing myCAFs in PDAC liver metastases interact with macrophages to suppress CD8⁺ T cell infiltration and activation.⁵⁴ Cross-cancer analyses further illustrate the immunomodulatory potential of myCAFs. Classically, myCAFs enforce immune evasion through physical mechanisms. By depositing dense collagen and fibronectin networks, myCAFs increase interstitial pressure and restrict T cell access to cancer nests.²⁰ More recently, LRRC15⁺ myCAFs have been identified as a universal response to high TGF- β signaling in the tumor bed.^{12,48} LRRC15⁺ myCAFs dampen intratumoral CD8⁺ T cell function and thus drive resistance to immune checkpoint blockade.²⁵ Other myCAF subsets exert distinct effects on the immune microenvironment. Senescent myCAFs (senCAFs) promote immunosuppressive macrophages, inhibit T cell and natural killer (NK) cell activation in PDAC and breast cancer through a matrisome-rich senescence-associated secretory phenotype (SASP).^{55,56} Moreover, residual ANTXR1⁺ myCAFs after chemotherapy inhibit

CD8⁺ T cell function in ovarian cancer.¹³ These observations highlight the diverse mechanisms by which myCAFs modulate tumor immunity and cancer progression across malignancies.

Antigen-presenting CAFs

The characterization of apCAFs marked a pivotal expansion in stromal immunobiology, moving the field beyond the binary view of fibroblasts as merely structural or inflammatory^{39,57} (Figure 1). Studies as early as the 1980s hinted that fibroblasts and other non-professional antigen-presenting cells could express MHC-II under interferon-gamma (IFN- γ) stimulation, with CIITA acting as the key transcriptional regulator.^{58–60} Single-cell studies in PDAC first identified apCAFs as a distinct CAF population expressing MHC-II genes and antigen presentation machinery (*H2-Aa*, *H2-Ab1*, and *Cd74* in mice; *HLA-DRA*, *HLA-DRB1*, *HLA-DPA1*, *HLA-DQB1*, and *CD74* in humans).^{39,57} While other CAF populations may transiently express MHC-II upon IFN- γ stimulation, apCAFs represent a transcriptionally distinct population with constitutively high MHC-II expression.^{8,11,48}

In PDAC, apCAFs are closely associated with mesothelial cells.³¹ apCAFs possess the machinery to engage CD4⁺ T cells but often lack costimulatory molecules (CD80 and CD86) required for effective priming, suggesting they may function as immune decoys that induce T cell tolerance via anergy or Treg differentiation.^{31,39} Since their discovery in PDAC, apCAFs have been detected across multiple cancer types with conflicting functional reports. apCAFs impede neoadjuvant chemoinmunotherapy in lung cancer,⁶¹ yet promote T cell activation and anti-tumor immunity in gastric cancer,⁶² and correlate with favorable outcomes in breast cancer.⁹ These discrepancies are partly clarified by recent lineage-resolved analyses showing two major apCAF origins: CD24⁺ mesothelial-like apCAFs and CD52⁺ hematopoietic-linked apCAFs.¹¹ CD52⁺ apCAFs likely represent recruited hematopoietic cells that have adopted a strong stromal gene signature, causing them to cluster with mesenchymal rather than immune cells in scRNA-seq datasets. This distinction suggests that C1q-mediated apCAF immunity in mouse lung cancer is largely attributable to the CD52⁺ subset, which predominantly produces C1q.^{11,63}

The two apCAF lineages occupy different spatial niches. CD24⁺ apCAFs localize near cancer cells, likely arise from mesothelial progenitors exposed to wound-related IL-1/TGF- β cues, and are enriched in peritoneal and chemoradiation-resistant niches where T cells often show exhausted or immunosuppressive features.^{11,31} In contrast, CD52⁺ apCAFs localize to lymphocyte-dense regions, suggesting possible roles in tertiary lymphoid structures (TLS) formation or maintenance, though evidence in murine models remains limited as most standard mouse tumors do not spontaneously form TLS.⁶⁴ Despite lineage differences, both populations upregulate SPP1, suggesting shared links to metastasis and therapy resistance.¹¹ To conduct more definitive studies, there is a critical need to define the intrinsic molecular regulators of apCAFs.

OTHER STROMAL POPULATIONS

While myCAF, iCAF, and apCAF represent major axes of functional polarization, single-cell atlases frequently identify additional mesenchymal clusters. A meta-analysis of the major

Table 1. Summary of CAF subtypes and marker signatures across major single-cell and spatial transcriptomic studies

Year	Reference	Tissue	Annotation	Key gene signature	Proposed conserved phenotype
2025	Chen et al. ¹¹	pan-tissue	ssICAF	<i>PI16, CFD, PLA2G2A, CD34, DPT</i>	universal progenitor
			iCAF	<i>CXCL8, CXCL2, IL6, LIF, HGF</i>	iCAF
			myCAF	<i>POSTN, FAP, ACTA2, MMP11, LRRC15, COL10A1</i>	myCAF
			CD24 ⁺ apCAF	<i>CD74, HLA-DRA, MSLN, CD24, KRT19, EZR</i>	apCAF
			CD52 ⁺ apCAF	<i>CD74, HLA-DRA, CD37, CD52, C1QA, C1QB, C1QC</i>	apCAF
			proliferative CAF	<i>MKI67, TOP2A, CENPF, UBE2C, STMN1</i>	proliferating CAF
			pericyte	<i>RGS5, MCAM, CSPG4, CD36, ACTA2</i>	mural
			smooth muscle cell	<i>MYH11, RERGL, ACTA2, PLN, DES</i>	mural
2024	Belle et al. ⁵⁵	pancreas	apCAF	<i>CD74, HLA-DRA, MSLN</i>	apCAF
			iCAF	<i>CXCL12, IL6, HAS1</i>	iCAF
			myCAF	<i>THY1, FAP, ACTA2</i>	myCAF
			senCAF	<i>CDKN2A, PLAUR</i>	myCAF
2024	Gao et al. ⁴⁸	pan-tissue	progenitor-like fibroblast	<i>PI16, CD34, APOD, COL15A1</i>	universal progenitor
			myofibroblast	<i>MMP11, COL10A1, POSTN, LRRC15, CTHRC1, HOPX, MMP1, WNT5A, ACTA2</i>	myCAF
			inflammatory fibroblast	<i>CXCL1, CXCL2, IL6, CLU, CTGF, HGF, HSPA6</i>	iCAF
			CTNNB1 ⁺ fibroblast	<i>WSB1, DDX17, CTNNB1</i>	unclassified
			proliferative fibroblast	<i>STMN1</i>	proliferating CAF
			antigen-presenting fibroblast	<i>CD74, HLA-DRA</i>	apCAF
			mesothelial cell	<i>KRT18, SLPI, MSLN</i>	mesothelial
			smooth muscle cell	<i>MYH11, ACTA2, TAGLN, HHIP</i>	mural
			pericyte	<i>RGS5, CD36, NOTCH3</i>	mural
			mCAF	<i>MMP11, POSTN</i>	myCAF
2023	Cords et al. ⁶⁵	breast	iCAF	<i>PI16, CFD, PLA2G2A, APOD, CD34</i>	universal progenitor
			vCAF	<i>MYH11, RERGL, ACTA2</i>	mural
			tCAF	<i>ENO1, GAPDH, NDRG1, MME</i>	iCAF
			hsp tCAF	<i>HSPH1, VEGFA</i>	iCAF
			rCAF	<i>CCL19, CCL21, IGFBP5, APOE</i>	FRC
			apCAF	<i>CD74, HLA-DRA, HLA-DRB1</i>	apCAF
			dCAF	<i>MKI67, STMN1, TOP2A, CENPF</i>	proliferating CAF
			ifnCAF	<i>CXCL9, CXCL10, CXCL11, IDO1</i>	iCAF
			Pericyte	<i>MCAM, RGS5, ACTA2</i>	Mural
			universal progenitor	<i>Pi16, Dpt, Cd55, Ly6c1</i>	universal progenitor
2022	Krishnamurty et al. ²⁵	pancreas	LRRC15 ⁺ CAF	<i>Lrrc15, Tagln, Acta2, Col8a1</i>	myCAF

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Table 1. Continued

Year	Reference	Tissue	Annotation	Key gene signature	Proposed conserved phenotype
2022	Grout et al. ²⁰	lung	PI16 ⁺ Fib	<i>PI16, LEPR, CD34, ADH1B</i>	universal progenitor
			Inf. Alv. Fib	<i>IL6, ICAM1, TCF21</i>	iCAF
			Alv. Fib	<i>ADH1B, TCF21</i>	universal progenitor
			CLU ⁺ Fib	<i>CLU, WIF1</i>	universal progenitor/iCAF
			Meso Fib	<i>PDGFRA, KRT8, KRT18, CD74, EZR, MSLN, SLPI</i>	mesothelial/apCAF
			ADH1B ⁺ CAF	<i>LEPR, ADH1B, TCF21</i>	universal progenitor
			FAP ⁺ CAF	<i>FAP, POSTN, LRRC15, CTHRC1</i>	myCAF
			FAP ⁺ α-SMA ⁺ CAF	<i>FAP, ACTA2, CTHRC1</i>	myCAF
2022	Foster et al. ⁶⁶	breast/pancreas	MYH11 ⁺ α-SMA ⁺ CAF	<i>WIF1, MYH11, ACTA2</i>	mural/myCAF
			SSL CAF	<i>Dpp4, Pi16, Cd34, Dpt</i>	universal progenitor
			MR CAF	<i>Postn, Lrrc15, Cdh11, Runx1, Thbs2, Fsp1</i>	myCAF
			IM CAF	<i>Il1r1, Cxcl1, Cxcl12, Il6, Ifngr1, Cd74, H2-Aa</i>	iCAF/apCAF
2021	Affo et al. ²⁸	liver	iCAF	<i>Cxcl12, Hgf, Il6,</i>	iCAF
			myCAF	<i>Col15a1, Col8a1, Fbn2, Postn, Vcan, Col6a3, Has2</i>	myCAF
2021	Hutton et al. ³⁸	pancreas	mesCAF	<i>Msln, Upk2b, Gpm6a, Upk1b</i>	mesothelial/apCAF
			CD105 ⁺ CAF	<i>Eng, Cspg4, Fap, Ly6c1, Postn, Cxcl14, Col8a1, Igfbp5</i>	universal progenitor/myCAF
			CD105 ⁻ CAF	<i>Cxcl2, Sod2, Gas1, Cd74, H2-Ab1, Msln, Krt18, Ezr, Cd14, Nos2</i>	iCAF/apCAF
2021	Buechler et al. ⁷	pan-tissue	pan-tissue adventitial	<i>Dpt, Pi16, Dpp4, Ly6c1, Cd34, Has1, Plin2</i>	universal progenitor
			pan-tissue parenchymal	<i>Dpt, Col15a1, Penk</i>	universal progenitor
			perturbed-state myofibroblast	<i>Lrrc15, Acta2, Col11a1, Cthrc1</i>	myCAF
2021	Wang et al. ⁶⁷	pancreas	iCAF	<i>APOD, C7, PTGDS</i>	universal progenitor/iCAF
			myCAF	<i>POSTN, COL10A1, MMP11, SDC1</i>	myCAF
			meCAF	<i>PLA2G2A, CRABP2</i>	iCAF
			apCAF	<i>CD74, HLA-DRA</i>	apCAF
2021	Xing et al. ⁶⁸	lung	iCAF	<i>CFD, CXCL14, CXCL12, GSN</i>	universal progenitor/iCAF
			CAF	<i>FAP, PDPN, TGFB1, POSTN</i>	myCAF
			apCAF	<i>SLPI, CD74, HLA-DRA, HLA-DRB1</i>	apCAF
			myofibroblast	<i>PTN, ACTA2, TAGLN, MYL9</i>	mural
2020	Dominguez et al. ¹²	pancreas	IL1 CAF	<i>IL1R1, IL6, LIF, CXCL1, HAS1</i>	iCAF
			TGF-β CAF	<i>LRRC15, ACTA2, CTHRC1, MMP11, TAGLN, SDC1, COL8A1, GJB2</i>	myCAF
2020	Kieffer et al. ¹⁹	breast	Ecm-myCAF	<i>ANTXR1, ASPN, COL3A1, THY1, SFRP2, COL10A1, LRRC15, COL6A3, LRRC17</i>	myCAF

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Table 1. Continued

Year	Reference	Tissue	Annotation	Key gene signature	Proposed conserved phenotype
2020	Wu et al. ⁶⁹	breast	TGF- β -myCAF	<i>ANTXR1, CST1, LOXL1, CST2, TGFB1, TGFB3, COL10A1, THBS4, THY1</i>	myCAF
			Wound-myCAF	<i>ANTXR1, SFRP4, CCDC80, OGN, DCN, SEMA3C, SFRP2, IGF1, IGF2</i>	myCAF
			Detox-iCAF	<i>ADH1B, C7, CXCL12, IL6, OGN, PLA2G2A, GPX3, PI16</i>	universal progenitor/iCAF
			IL-iCAF	<i>ITM2A, CXCL12, C7, SCARA5, DLK1, MEG3, RGMA, IL6</i>	universal progenitor/iCAF
			IFN- γ -iCAF	<i>CCL19, CXCL13, CXCL9, CCL2, IL34, CXCL10, CCL8, CCL5</i>	iCAF
			iCAF	<i>IGF1, CXCL12, FIGF, DLK1, CXCL1, IGF2, ALDH1A1, CD34</i>	universal progenitor/iCAF
			myCAF	<i>FN1, MMP11, COL12A1, COL1A1, PDPN, LOX, MMP2, FAP</i>	myCAF
			dPVL	<i>MYH11, ACTA2, MYL9, TAGLN, CD9, MYLK, MCAM, RGS5</i>	mural
			imPVL	<i>CD36, RGS5, NOTCH3, ITGA1, MCAM, MYLK, TAGLN</i>	mural
			pCAF	<i>Pdpn, Cxcl12, Saa3, Acta2, Fbn1, Cxcl1, Il6, Ly6c1</i>	iCAF/myCAF
2020	Friedman et al. ⁹	breast	sCAF	<i>S100a4, H2-Aa, Hspd1, Spp1</i>	apCAF
			NMF	<i>Gsn, Ppap2b, Ogn, Timp2, Igfbp6</i>	universal progenitor
			endothelial cell	<i>Pecam1</i>	endothelial
			pericyte	<i>Rgs5</i>	mural
			iCAF	<i>Dpt, Ifi205, Gsn, Figf, Svep1, Plpp3, Scara3, Pcolce2, Clec3b, Efemp1, Ifi27L2a, Col14a1, Ly6a, Ly6c1, Adamts5, Tnxb, C4b, C3</i>	iCAF
			myCAF	<i>Serpine2, Col15a1, Tnc, Sparcl1, Spp1, Tgfb1, Crf1, Thy1, Tagln, Col12a1, Col1a1, Cxcl14, Sdc1, Cthrc1, Col8a1, Thbs2, Acta2</i>	myCAF
			apCAF	<i>Cd74, H2-Ab1, Slpi, Ptgis, Nkain4, Hspb1, Cav1, Arhgdib, Lgals7, Spint2, Dmkn, Saa3, Clu, Slc9a3r1, Krt8, Cxadr, Ezr, Fth1</i>	apCAF
2019	Elyada et al. ³⁹	pancreas			

studies summarized in this review (Table 1) reveals that many clusters map to conserved biological identities. Recognizing a universal CAF framework with canonical categories and nomenclature is essential to converge on true functional states, avoid over-classifying the TME, and advance the field.

Other universal progenitor-associated CAFs

Across studies, PI16⁺/DPT⁺/CD34⁺ universal progenitor-like clusters are frequently detected alongside polarized CAF states.^{7,66} These cells represent a quiescent state lacking the robust contractile programs of activated CAFs. They likely

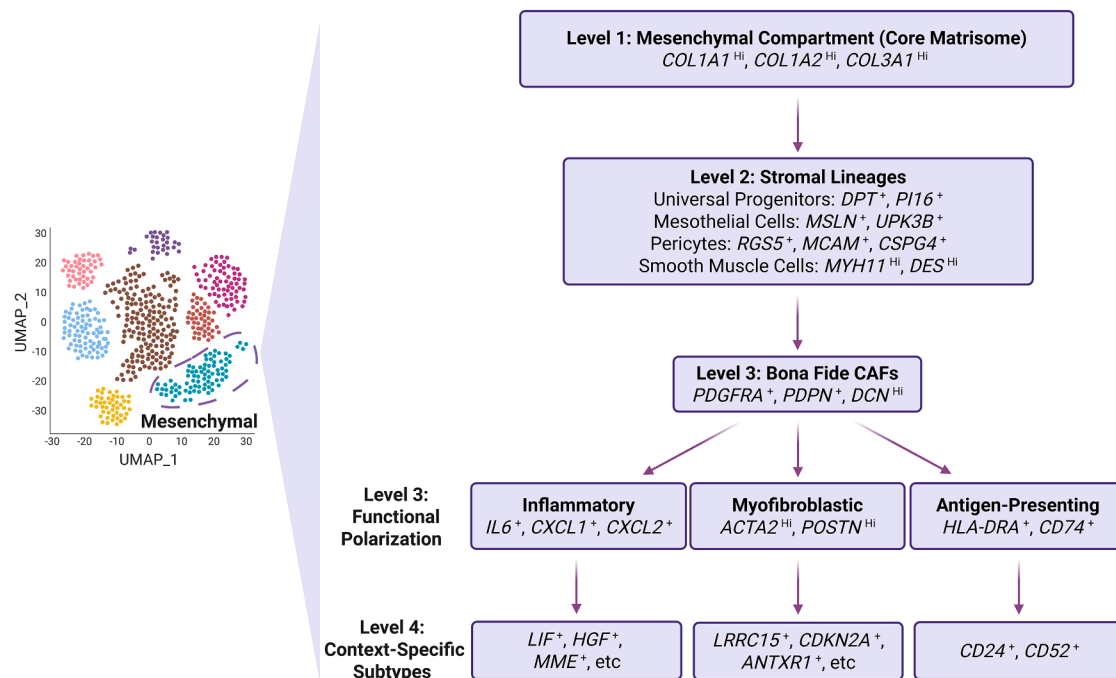


Figure 2. A hierarchical classification system for CAFs

Left image shows a representative UMAP projection of the mesenchymal compartment from single-cell transcriptomic data. Right image presents the proposed four-level classification hierarchy. Level 1 establishes the mesenchymal compartment through high expression of core matrisome genes (*COL1A1*, *COL1A2*, and *COL3A1*), broadly distinguishing stromal cells from epithelial, endothelial, and immune populations. Level 2 delineates distinct stromal lineages using lineage-defining markers: universal progenitors (*DPT*⁺ and *PI16*⁺), mesothelial cells (*MSLN*⁺ and *UPK3B*⁺), pericytes (*RGS5*⁺, *MCAM*⁺, and *CSPG4*⁺), and smooth muscle cells (*MYH11*^{Hi} and *DES*^{Hi}). Level 3 verifies bona fide CAF identity through expression of *PDGFRA*, *PDPN*, and high *DCN*, then resolves CAFs into three conserved functional polarization states: inflammatory (*IL6*⁺, *CXCL1*⁺, and *CXCL2*⁺), myofibroblastic (*ACTA2*^{Hi} and *POSTN*^{Hi}), and antigen-presenting (*HLA-DRA*⁺ and *CD74*⁺). Level 4 further dissects these broad categories into context-specific subtypes driven by local signals, interventions and cellular states, including *LIF*⁺, *HGF*⁺, and *MME*⁺ iCAFs; *LRRC15*⁺, *CDKN2A*⁺, and *ANTXR1*⁺ myCAFs; and *CD24*⁺ and *CD52*⁺ apCAFs.

represent a local reservoir of naive tissue-resident fibroblasts that have not yet been fully rewired by tumor-derived cues. Consequently, many new CAF clusters described in the literature may represent this conserved progenitor pool captured at various stages of activation.⁶⁷ Future studies should determine whether this transition involves discrete states or spatially defined niches, as universal progenitors differentiate into mature CAF phenotypes.

Mural cells: Pericytes and smooth muscle cells

A significant source of confusion in stromal biology arises from the frequent inclusion of mural cells within “CAF” clusters. While mural cells, comprising pericytes and smooth muscle cells (SMCs), share broad mesenchymal similarity with fibroblasts, they represent functionally and transcriptionally distinct entities. In many single-cell studies, mural cells are misclassified as myCAFs often because they express the contractile marker α -SMA (*ACTA2*). However, high-resolution profiling reveals critical distinctions. Unlike bona fide myCAFs, mural cells typically lack canonical fibroblast markers, such as *PDGFR α* or podoplanin (*PDPN*).^{11,12} Instead, they express distinct transcriptional programs: pericytes express *RGS5*, *MCAM*, and *CSPG4*, while SMCs express *MYH11* and *DES*.^{18,51,69} Furthermore, mural cells express significantly higher levels of α -SMA than myCAFs, reflecting their primary role in vascular tone and contractility rather than matrix remodeling. Recent evi-

dence suggests mural cells may also serve as potential progenitors for specific fibroblast subsets. For instance, in lung cancer, mural cells can contribute to the formation of CCL19⁺ fibroblastic reticular cells (FRCs) and thus lymphoid structures.³³ Similarly, in breast cancer, the pericyte-associated cluster defined as vascular CAFs (vCAFs) strongly align with this lineage, displaying a distinct perivascular gene signature that distinguishes them from myCAFs.^{65,70} Therefore, distinguishing CAFs from mural cells will advance the field by elucidating their spatial distributions, plasticity, and function within the TME.

Proliferating CAFs

Finally, most high-fidelity tumor stromal atlases identify a distinct cluster of fibroblasts that highly express cell-cycle genes, including *MKI67*, *TOP2A*, *STMN1*, and *CDK1*.^{11,25,48} While these “proliferating CAFs” are often clustered separately due to the dominance of cell-cycle transcripts over functional gene signatures, they may represent an expanding fraction of CAF populations. Notably, transcriptional analyses indicate that the cluster coined proliferating CAFs often retain underlying gene signatures associated with myCAFs.^{11,12} This observation has led to the hypothesis that, in many contexts, myCAFs may represent the most actively proliferating population within the tumor stroma, driving the expansion of the desmoplastic reaction.

Stromal phenotypes beyond cancer

The conserved CAF phenotypes described previously may not be unique to the TME. Accumulating evidence suggests that the major fibroblast programs observed in cancer are shared across pathological conditions. Cross-tissue studies spanning healthy, inflammatory, fibrotic, and cancerous states have revealed that non-cancer pathologies drive consistent shifts in fibroblast composition and phenotypes.⁴⁸ Inflammatory fibroblasts bearing transcriptional resemblance to iCAFs have been described in rheumatoid arthritis, inflammatory bowel disease, and other chronic inflammatory conditions,^{71,72} while myofibroblastic programs are hallmarks of idiopathic pulmonary fibrosis, systemic sclerosis, and liver cirrhosis.^{73–75} MHC-II-expressing fibroblasts with antigen-presenting features have likewise been observed in inflamed joints and intestinal tissue.^{76,77} These parallels reinforce the concept that tumors may co-opt conserved tissue damage-response programs rather than generating entirely novel stromal states, and suggest that insights from fibrosis and inflammatory disease research may inform fibroblast biology in cancer and vice versa.

CAF classification framework

Given the inherent complexity of the stromal compartment and the rapid accumulation of high-dimensional stromal datasets, a consensus taxonomy is becoming increasingly essential to integrate findings across different tumor types and technical platforms. To facilitate this synthesis and help resolve the complex overlap between developmental origins and functional states based on current evidence, we propose a hierarchical four-level marker framework as a conceptual guide (Figure 2).

Level 1: Establishing the mesenchymal compartment

The foundational step in any stromal analysis should be to broadly distinguish the mesenchymal compartment from the epithelial, endothelial, and immune components of the TME. Regardless of their specific lineage or function, all stromal cells are unified by their capacity to robustly express ECM genes. Therefore, the primary gate can focus on high expression of fibrillar collagen genes, such as *COL1A1*, *COL1A2*, and *COL3A1*, which are typically sufficient to identify the major mesenchymal compartment. Critically, the subsequent exclusion of non-stromal cells should be applied with nuance to preserve lineage-transitioning populations. While bona fide epithelial cells and canonical immune effectors should be rigorously removed, strict filtering based on cytokeratins (*KRT*) or CD45 (*PTPRC*) alone may inadvertently exclude biologically significant subsets. For instance, mesothelial-associated cells often express epithelial genes such as cytokeratin genes, while hematopoietic progenitors inherently express *PTPRC* (CD45). Notably, in unsupervised analyses, these progenitors typically co-cluster with the mesenchymal compartment despite their mixed lineage signatures, likely driven by the dominance of their ECM transcriptional program. Thus, level 1 defines the “ECM-competent” landscape, purposefully retaining these ambiguous populations for precise resolution in level 2.

Level 2: Delineating stromal lineages

The second level of resolution aims to delineate distinct stromal cell lineages. Universal fibroblast progenitors are characterized by expression of *PI16* or *DPT*.⁷ The mural lineage can be resolved into pericytes (marked by *RGS5*, *MCAM*, and *CSPG4*)

and smooth muscle cells (defined by high *MYH11* and *DES* expression).^{11,51} Normal tissue-resident mesothelial cells are distinguished by *MSLN*, *UPK3B*, and *KRT19*, while potential hematopoietic-derived progenitors retain detectable levels of *PTPRC*.¹¹

Level 3: Defining CAF identity and broad functional polarization

The third level first aims to verify CAF identity. *PDGFRA* expression establishes a foundational fibroblast identity: under normal conditions, resting universal fibroblasts (*PI16*⁺ or *DPT*⁺) are the major cell types expressing *PDGFRA*, while non-fibroblast progenitors such as mesothelial cells and mural cells are typically *PDGFRA* negative.^{7,11,12,31,51} However, these progenitor populations can acquire *PDGFRA* expression in tumor settings, reflecting their transition toward a CAF identity.^{11,31} CAF identity can be further supported by *DCN* and *PDPN* expression, as CAFs are typically *DCN*-high and *PDPN*-positive^{39,78}; however, *PDPN* is not CAF-specific and can also be expressed by lymphatic endothelial cells and normal mesothelial cells.^{12,31}

The TME then imposes functional programs that drive these cells into broad polarization states (iCAFs, myCAFs, and apCAFs) observed across most solid tumors. Of note, iCAFs typically retain higher *PDGFRA* expression compared to myCAFs or apCAFs.^{11,31,39} In addition, iCAFs are defined by an inflammatory cytokine/chemokine transcriptional program, including *IL-6*, *CXCL1*, and *CXCL2*. myCAFs are characterized by genes indicating activation of the contractile cytoskeleton and robust ECM deposition, including increased expression of *ACTA2* and *POSTN*. Notably, *FAP* gene expression likely marks an early activation or “primed” state during the transition from universal progenitors toward CAF differentiation; accordingly, *FAP*⁺ CAFs may represent a heterogeneous population spanning the progenitor-to-iCAF/myCAF continuum, with myCAFs exhibiting the highest expression levels.⁸ Lastly, apCAFs are identified by robust expression of MHC-II machinery (*HLA-DRA*, *CD74*).

Level 4: Context-specific subtypes

The final level resolves broad CAF states into specialized subtypes shaped by defined biological contexts. Here, “context” refers to four intersecting axes: tissue or organ of origin, local spatial niche, disease stage, and perturbation state, including therapy and other experimental or pharmacological interventions. These subtypes should therefore be interpreted as context-dependent refinements of the major conserved CAF phenotypes rather than as entirely separate CAF lineages. This layer remains dynamic as new studies emerge, but several recurrent examples illustrate its utility.

myCAF subtypes may stratify by activation intensity and spatial context. LRRC15⁺ myCAFs represent the most consistently identified subtype, marking a highly activated, TGF- β -driven state associated with T cell exclusion and immune evasion.^{12,25} CDKN2A⁺ myCAFs represent a senescent variant exhibiting a senescence-associated secretory phenotype, likely arising following therapy-induced damage or prolonged activation.^{55,56} Alongside this, *ANTXR1* has been identified as a broader lineage marker that captures a dynamic spectrum of myCAFs.¹⁹

iCAF subtypes diverge based on dominant cytokine output and function. LIF⁺ and HGF⁺ iCAFs represent the most frequently

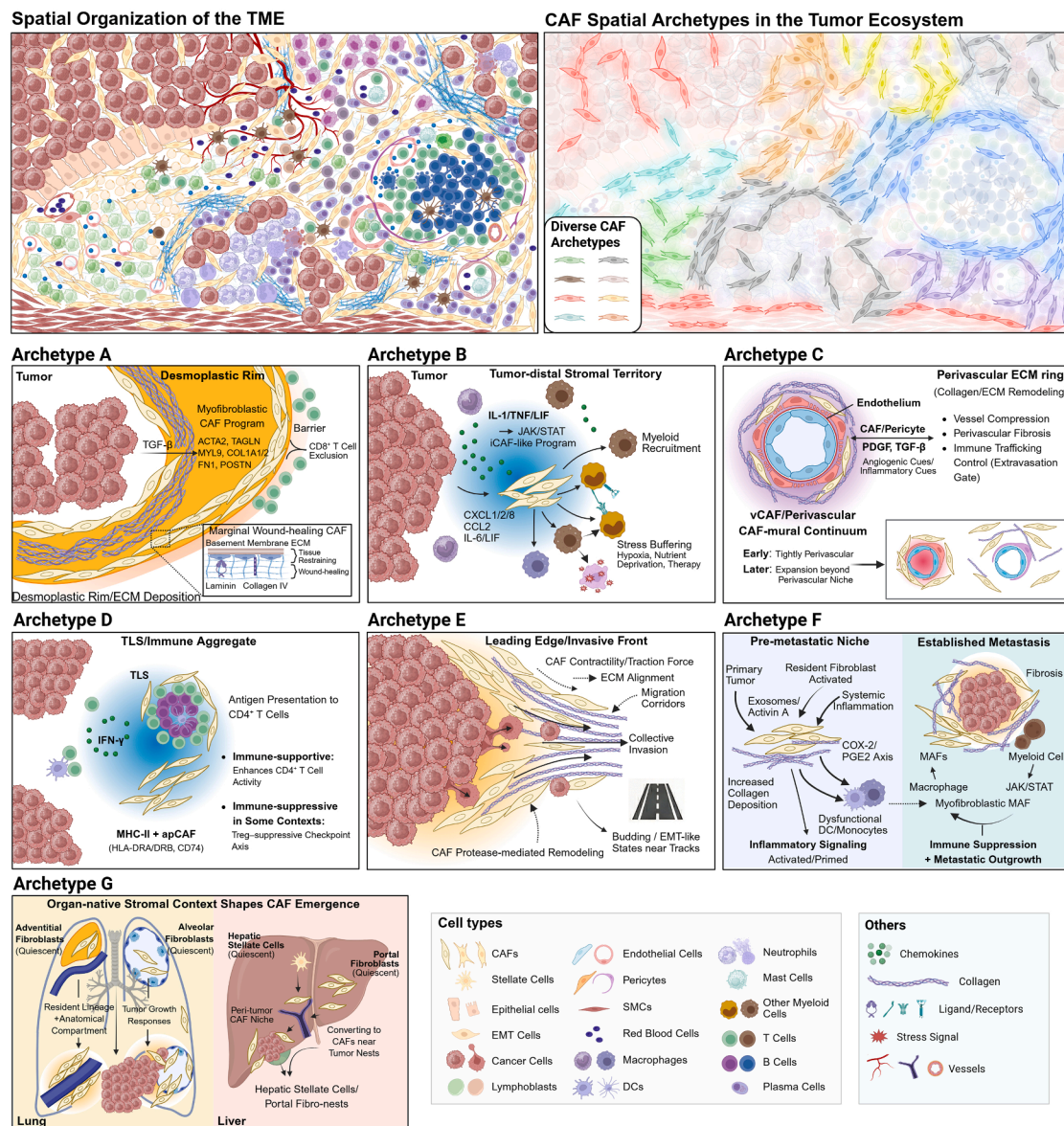


Figure 3. Spatial archetypes of CAFs in the tumor ecosystem

Upper images overview the spatial organization of the TME (left) and distribution of CAF archetypes within the tumor ecosystem (right). Lower images depict seven recurring spatial archetypes. Archetype A (desmoplastic rim): TGF- β -driven myofibroblastic programs form ECM-dense barriers excluding CD8⁺ T cells, with a marginal wound-healing compartment that may restrain tumors. Archetype B (tumor-distal inflammatory stroma): IL-1/TNF/LIF-driven JAK/STAT-dependent iCAF programs recruit myeloid cells and buffer metabolic stress. Archetype C (perivascular compartment): a CAF-mural continuum around microvessels exchanges PDGF and TGF- β signals, influencing vessel compression, perivascular fibrosis, and immune trafficking. Archetype D (TLS/immune aggregate): IFN- γ -driven MHC-II⁺ apCAFs engage CD4⁺ T cells with dual outcomes depending on lineage and context, either supporting anti-tumor immunity or promoting Treg differentiation. Archetype E (invasive front): CAF contractility aligns ECM into migration corridors facilitating collective invasion alongside budding/EMT-like tumor states. Archetype F (pre-metastatic and metastatic niches): tumor-derived systemic signals prime organ-resident fibroblasts before colonization; established metastases develop myofibroblastic MAFs through macrophage-fibroblast JAK/STAT crosstalk. Archetype G (organ-specific context): native stromal architecture constrains CAF emergence, illustrated by lung (adventitial and alveolar fibroblasts in distinct compartments) and liver (stellate and portal fibroblasts converting near tumor nests).

annotated subpopulations, differing in their specific growth factor secretion profiles.^{11,48} Additionally, CD10 (*MME*)⁺ iCAFs, identified in breast cancer, constitute a functionally distinct inflammatory subtype characterized by high IL-6/IL-8 secretion.⁴⁷ This population provides a survival niche for cancer stem cells and promotes chemoresistance.

apCAF subtypes are most meaningfully resolved through their distinct signatures. CD24⁺ apCAFs are associated with mesothelial-like cells, where they have been implicated in regulatory T cell induction. Conversely, CD52⁺ apCAFs, of potentially hematopoietic origin, express complement genes and associate with lymphoid structure formation. In lower-resolution datasets,

these two populations may co-cluster into a single apCAF group, obscuring their origins.

CAF SPATIAL ORGANIZATIONAL PATTERNS

CAFs are continuously shaped by spatial cues, local ECM mechanics, and ligand gradients, yet they also actively remodel their surrounding milieu. CAF phenotype is therefore best understood as context-dependent programs operating within defined spatial niches, emerging from four major classes of spatially patterned inputs: diffusible signaling gradients, contact-dependent cues, tissue mechanics/ECM architecture, and metabolic conditions. Recent pan-cancer spatial multi-omics analyses validate this concept, demonstrating that conserved CAF-centered organizational patterns recur across cancers, each spatially coupled with distinct neighboring-cell contexts, interaction networks, tumor immune phenotypes, and clinical outcomes.¹⁰

Here, we use the term “spatial archetypes” to describe a recurring spatial-organizational and functional pattern of CAF localization, interaction, and microenvironmental association (Figure 3). These archetypes can be viewed as CAF-centered spatial configurations within the tumor ecosystem. A spatial niche refers to a local microenvironmental context, whereas a spatial archetype captures a broader recurring organizational motif that may encompass one or more niches. Accordingly, spatial archetypes represent higher-order organizational patterns rather than rigid taxonomic categories and may encompass multiple CAF states, neighborhood structures, and interaction modules that together shape integrated TME function. To facilitate cross-tumor comparison and reproducible application of this framework, we summarize the defining spatial relationships, dominant CAF programs, niche composition, and functional interpretation of each archetype (Table 2).

Archetype A (tumor-stroma interface/desmoplastic rim)

Archetype A comprises CAF programs enriched at the tumor-stroma interface, where malignant cells, fibroblasts, immune cells, and ECM assemble into a spatially organized compartment, acting as a hub that regulates tumor-stroma communication, matrix organization, and immune access.^{20,79–81} Interface-associated CAFs often display myofibroblastic, matrix-remodeling, and contractile programs, including *ACTA2*, *TAGLN*, *MYL9*, *COL1A1/2*, *FN1*, *POSTN*, and *TNC*, and in some settings *LRRC15/COL11A1/COL12A1*-associated signatures.^{10,12,20,48,82} Their local abundance and organization can shape immune infiltration patterns and influence response to immunotherapy, contributing to a structural, mechanical, and signaling barrier that increases tumor-immune separation.^{79–81} Notably, a study in human lung tumors showed that T cells actively migrate along aligned fibronectin and collagen fibers. However, fiber orientation guides them circumferentially around tumor islets rather than radially into them.⁷⁹ Thus, boundary ECM may constrain T cell positioning by redirecting migration, rather than by acting simply as a static wall.

TGF- β is a central and often dominant regulator in this archetype.^{25,48} In PDAC, TGF- β signaling is required for the development of *LRRC15*⁺ myCAFs,²⁵ which directly suppress CD8⁺ T cell function and diminish anti-PD-L1 response. In lung tumors, distinct peritumoral α -SMA⁺ layers, including

MYH11⁺ α -SMA⁺ SMCs and FAP⁺ α -SMA⁺ myCAFs, localize around tumor nests, correlate with reduced intratumoral T cell infiltration, and deposit a dense fibrillar matrix enriched in collagen XI and XII.^{20,82} In line with this, the TGF- β -driven myCAF cluster is detected at the junction between the tumor nest and the juxta-tumor compartment in breast cancer.¹⁹ Importantly, boundary-localized CAFs are not uniformly tumor-promoting. Spatial studies in HCC identified distinct fibroblast programs across the tumor core and margin, including a marginal wound-healing compartment characterized by basement membrane-associated ECM (laminins, collagen IV) and pro-fibrotic receptor features associated with tumor-restraining behavior, alongside an intratumoral inflammatory FAP⁺ compartment linked to worse outcomes.⁸³ Thus, archetype A is best viewed as a recurring spatial-organizational pattern that includes functionally divergent CAF programs. Depending on cancer type, stromal architecture, immune context, and other tissue components, these programs may reinforce immune exclusion or support tumor-restraining functions.

Archetype B (tumor-distal inflammatory stroma/myeloid-rich compartment)

Archetype B refers to CAF programs preferentially enriched in stromal territories distal from tumor nests and frequently co-localized with myeloid-rich regions, although similar programs may also extend into peritumoral stroma. A canonical model emerged from PDAC, in which IL-1- and JAK/STAT-dominant inflammatory cues reprogram tumor-distal fibroblasts toward an iCAF-like state, establishing a broader principle: CAF phenotype can be spatially regulated by the dynamic balance between diffusible inflammatory cytokines and contact- or mechanics-associated inputs rather than existing as a static state.^{15,16,39} Archetype B CAFs frequently express chemokines and cytokines (CXCL1/2/8, CCL2, IL-6, and LIF) that recruit macrophages and neutrophil-like populations and sustain immunosuppressive or tumor-promoting myeloid states in a cancer-type-dependent manner.⁴¹ Pan-cancer spatial analyses confirm that inflammatory CAF-centered niches are commonly enriched for macrophages, consistent with a coordinated stromal-myeloid ecosystem.¹⁰ For instance, this archetype has been defined in breast cancer, where iCAF clusters form distal inflammatory niches with tissue-resident FOLR2⁺ macrophages.¹⁹

Beyond stromal-myeloid coupling, archetype B CAFs can establish inflammatory paracrine circuits that help cancer cells adapt to metabolic and therapeutic stress. For example, CAF-derived IL-6 and LIF can activate JAK/STAT signaling in cancer cells to support survival under hypoxia, nutrient limitation, and treatment pressure; IL-6/IL-8-secreting CD10⁺GPR77⁺ CAFs sustain cancer stemness and chemoresistance; and therapy-induced iCAFs in rectal cancer produce survival-promoting secretome after chemoradiation.^{46,47,84,85} Thus, in contrast to archetype A, archetype B is better understood as a diffuse inflammatory-paracrine compartment that links cytokine-driven cancer-cell survival with therapy adaptation. At the ecosystem level, these two archetypes may act in concert: archetype A establishes the structural edge of the tumor while archetype B extends inflammatory, myeloid, and stress-adaptive signaling throughout the surrounding stroma, together maintaining an immune-excluded TME and promoting therapy resistance.

Table 2. Features of CAF spatial archetypes

Archetype	Functional readout/ translational relevance	Spatial relationship	Representative CAF programs/markers	Niche composition	Representative refs
A. Tumor-stroma interface/ desmoplastic rim	matrix organization, immune exclusion or stromal containment, TGF- β signaling, and immunotherapy response/ resistance.	cancer-proximal CAF programs enriched at the cancer-stroma interface, surrounding cancer nests/ islets or marginal stromal compartments.	myCAF/ECM-remodeling: ACTA2, TAGLN, MYL9, COL1A1/2, FN1, POSTN, TNC; context-dependent LRRC15, COL11A1, COL12A1.	cancer cells, cancer- proximal myCAFs, dense ECM, excluded CD8 ⁺ T cells, Tregs, macrophages	10,12,19,20,25,48,79–83
B. Tumor-distal inflammatory stroma/myeloid-rich compartment	myeloid recruitment, cytokine-driven cancer-cell survival, stress adaptation, and therapy resistance/ adaptation.	tumor-distal or peritumoral stromal territories, often away from direct cancer-cell contact; similar programs may extend into peritumoral stroma.	iCAF/stress-response: IL6, LIF, CXCL1/2/8, CCL2, HGF, CXCL12; IL-1/TNF/OSM/ JAK/STAT-associated programs.	iCAFs, macrophages, neutrophil-like populations, stressed cancer cells	10,15,16,19,39,41,43,46,47,84,85
C. Perivascular mesenchymal compartment/ CAF-mural Continuum	vascular integrity, perfusion, vessel compression, hypoxia, drug accessibility, and immune trafficking.	mesenchymal programs organized around microvessels and vessel walls.	perivascular/mural- associated: PDGFRB, ACTA2, RGS5, MCAM, CSPG4, MYH11, DES	endothelial cells, pericytes, smooth muscle cells, vascular-associated CAFs, and perivascular immune cells.	33,51,65,86–91
D. Immune-adjacent antigen-presenting Stroma	T cell activation or Treg induction; TLS-associated immunity.	CAF programs adjacent to lymphocyte-rich stromal territories, TLS-like aggregates, or cancer/ serosal immune interfaces.	apCAF: HLA-DRA, HLA- DRB1, HLA-DPA1, HLA- DQB1, CD74, CIITA; IFN- γ associated programs.	apCAFs, CD8 ⁺ T cells, CD4 ⁺ T cells, Tregs, B cells/TLS components, macrophages/ DCs; cancer cells in mesothelial niches.	11,31,61–63
E. Invasive front/leading- edge matrix- remodeling CAFs	directional cancer-cell migration, collective invasion, stromal tracks, and metastatic recurrence risk.	CAFs at invasive fronts, tumor-budding zones, and aligned ECM bundles; co- localized with invading cancer cells.	matrix-remodeling/ contractile: FN1, POSTN, COL1A1/2, TNC, ACTA2; MMP/LOX-related and YAP mechanotransduction programs.	invading cancer cells, leading-edge myCAFs, aligned fibronectin/collagen tracks, and macrophages in some settings.	50,53,92–95
F. Pre-metastatic and metastatic stroma	pre-metastatic priming, immune suppression, metastatic outgrowth.	fibroblast states in pre- metastatic organs or established metastatic lesions, shaped by target- organ stromal architecture.	fibrotic programs of metastatic sites: COL1A1/2, POSTN, VCAN, SPP1, ACTA2; JAK/STAT- associated macrophage- fibroblast programs.	organ-resident fibroblasts, disseminated cancer cells, macrophages, endothelial cells, and organ-specific immune cells.	54,96–99
G. Organ-specific stromal context	explains tissue-specific CAF abundance, neighborhood partners, clinical associations, and organ- specific vulnerabilities.	CAF programs constrained by the native anatomical scaffold: resident progenitor reservoirs, compartmental boundaries, ECM composition, perfusion, and immune-entry routes.	organ-dependent reservoirs: e.g., lung adventitial/ alveolar/myofibroblast-like fibroblasts; liver stellate and portal fibroblast programs	resident fibroblasts, resident macrophages, vascular/ perfusion structures, organ- specific ECM	7,10,11,26,29,51,65,68,75,83,100–103

Archetype C (perivascular mesenchymal compartment/CAF-mural continuum)

Archetype C refers to mesenchymal programs organized around microvessels and vascular walls, where fibroblast-like states frequently overlap with mural lineages (pericytes and vascular SMCs). This region presents a recurring challenge for spatial profiling: *ACTA2/PDGFRB/DES*-high cells may reflect pericytes or SMCs rather than canonical CAFs, depending on vessel caliber, tissue context, and assay resolution. Recent spatial analytical studies have improved delineation of this niche through mural-aware annotation strategies that integrate lineage markers, vessel proximity, and tissue morphology.^{51,86}

Archetype C represents a vascular-adjacent ecosystem in which endothelial cells, mural cells, and CAFs exchange PDGF, TGF- β , angiogenic, inflammatory, and ECM-remodeling signals.^{87,88} Functionally, this ecosystem shapes vascular integrity and perfusion through perivascular ECM remodeling and mural-like contractile programs, regulates immune trafficking through endothelial-stromal organization and perivascular barriers, and influences hypoxia and drug accessibility through vessel compression and altered interstitial transport. In breast cancer, a vCAF state identified in close association with microvessels in early lesions expands beyond the perivascular niche during tumor progression, and its signature is linked to increased metastatic risk.^{65,89} Furthermore, these perivascular populations do not act in isolation; they may cooperate with FAP⁺ myCAFs to synergistically drive metastatic dissemination.⁹⁰ In lung cancer, pericytes can give rise to CCL19⁺ FRCs that support lymphoid structure formation and lymphocyte trafficking.³³ Clinically, vessel-associated stromal readouts linked to vascular perfusion, hypoxia, and immune trafficking may offer more mechanistically informative biomarkers than generic CAF signatures.⁹¹

Archetype D (immune-adjacent antigen-presenting stroma)

Archetype D encompasses the spatially organized programs predominantly localized within lymphocyte-rich stromal territories, including lymphoid aggregates and TLS-adjacent regions. A defining feature of this spatial niche is the active engagement of antigen-presentation machinery, likely via the IFN- γ -CIITA axis.¹¹

Archetype D represents a dynamic immune-interface whose function is dictated by the precise spatial arrangement and cellular origin. In lung tumors, CD52⁺ apCAFs spatially support local CD4⁺ T cell immunity through *in situ* antigen presentation and C1q-C1qbp-mediated survival signaling, where fibroblast-specific perturbation of MHC-II or C1q actively impairs antitumor immunity.⁶³ Similarly, in gastric cancer, apCAFs preferentially enriched near TLS promote T cell activation and cytotoxicity, favor pro-inflammatory macrophage polarization, and correlate with immune-responsive features.⁶² Conversely, due to a potential mesothelial cell of origin, CD24⁺ apCAFs establish an alternative spatial configuration, positioning themselves directly adjacent to both cancer and immune cells at serosal interfaces.^{11,31} In this specific spatial context within PDAC, apCAFs have been shown to utilize MHC-II programs to promote antigen-specific CD4⁺ T cell conversion toward Tregs.³¹ In neo-adjuvant chemo-immunotherapy treated non-small cell lung cancer (NSCLC), IFN- γ can drive the expansion of apCAF-like

states within localized fibrotic hotspots.⁶¹ Within these specific treatment-induced niches, apCAFs are linked to impaired CD4⁺ T cell function. Collectively, archetype D is best understood as a contextual immune-stromal interface likely shaped by local interferon tone, lineage origin, and surrounding immune architecture.

Archetype E (invasive front/leading-edge matrix-remodeling CAFs)

Archetype E comprises CAF programs enriched at the invasive front, tumor budding zones, and along aligned ECM bundles that serve as structural corridors for tumor dissemination. In contrast to the boundary-restricted features of archetype A, archetype E is defined by active remodeling at the leading edge, where myCAFs co-localize with invading cancer cells and generate migration-permissive microenvironments through force-mediated remodeling, protease activity, and matrix reorganization.^{92,93} Cancer cells at the invasive front often correspondingly upregulate ECM organization and focal-adhesion programs that facilitate navigation along these stromal tracks. Mechanistically, CAFs align fibronectin into anisotropic matrices through contractility and traction-force-dependent pathways, directly promoting directional cancer cell migration at human tumor-invasion fronts.⁹⁴

Recent work supports a mechanofeedback loop in which stiffness-responsive CAF signaling amplifies invasion-track formation: in breast cancer, matrix stiffening activates YAP-dependent periostin production in CAFs, promoting collagen crosslinking and collective invasion, with these programs correlating with higher metastatic recurrence risk when coupled with high collagen abundance.^{50,95} Together with aggressive cancer cell states such as budding and EMT-like programs, these findings indicate that metastatic risk is shaped not only by tumor-intrinsic factors but also by stromal architecture at the invasive front.⁵³

Archetype F (pre-metastatic and metastatic stroma)

Archetype F captures fibroblast states that emerge during pre-metastatic niche formation and further diversify after metastatic lesions are established. These stromal states do not simply mirror primary tumor CAFs; they are built on the target organ's native mesenchymal architecture (lung adventitial fibroblasts, liver stellate, and portal fibroblasts) and reshaped by tumor-derived systemic signals, inflammatory circuits, and local immune-vascular interactions. A staged model is emerging in which organ-resident fibroblasts are first primed before overt metastasis and later diversify into metastasis-associated fibroblast (MAF) states that coordinate fibrosis, immune suppression, and metastatic outgrowth.^{54,96,97}

Evidence for early, organ-specific fibroblast priming is accumulating. In lung pre-metastatic niche models, COX-2-expressing adventitial fibroblasts organize a metastasis-permissive niche through PGE2-driven dendritic cell (DC) dysfunction and suppressive monocyte states; fibroblast-specific *Ptgs2* deletion reduced metastasis and improved anti-PD-1/DC-vaccine response.⁹⁶ In parallel, breast cancer-derived Activin A induced profibrotic programs and collagen deposition in lung fibroblasts at pre-metastatic stages, and genetic targeting of Activin A attenuated lung metastasis.⁹⁷ In established metastatic lesions, MAF states remain

strongly shaped by organ context. In PDAC liver metastasis, macrophage-fibroblast JAK/STAT signaling drives immunoregulatory myCAFs whose reciprocal outputs promote tumor growth and immune suppression.⁵⁴ In colorectal liver metastasis, a VCAN-enriched CAF state co-localizes with myofibroblast-like CAFs and SPP1⁺ macrophages in tumor fibrotic regions, consistent with reproducible fibroblast-myeloid-fibrosis neighborhoods supporting metastatic outgrowth.⁹⁸ Tumor-derived exosomal miR-188-3p can also activate hepatic stellate cells before metastasis, suggesting that pre-metastatic stromal changes may be detectable and potentially actionable as early biomarkers or interception targets.⁹⁹

Archetype G (organ-specific stromal context)

Archetype G captures the organ-native stromal template that shapes how CAF states are generated and spatially deployed. Its defining structure is the pre-existing anatomical scaffold of each tissue: resident fibroblast reservoirs, compartmental boundaries, ECM composition, vascular and perfusion architecture, and immune-entry routes.^{7,10,51} These features determine which precursors are available, where cancer-derived ligands and mechanical cues are encountered, and how otherwise conserved CAF programs assemble into organ-specific neighborhoods.^{7,10} In this context, CAF marker expression is most informative when interpreted together with anatomical position and lineage context. Examples include alveolar versus adventitial lung stroma, perivascular or serosal compartments, and perisinusoidal or portal liver stroma.^{11,26,29,65,100} This may explain why pan-cancer CAF spatial programs can recur across malignancies while differing in abundance, neighboring-cell partners, and clinical associations across tissues.¹⁰

Examples from lung and liver illustrate this archetype. In NSCLC, adventitial, alveolar, and myofibroblast-like fibroblasts occupy distinct native lung niches; tumor remodeling appears to activate these tissue-resident reservoirs through inflammatory, stress-response, and fibrillar collagen programs, with alveolar/adventitial features associated with more favorable prognosis and myofibroblast enrichment with worse outcome.^{68,100} POSTN⁺ CAFs represent a pathologically remodeled lung CAF state that co-localizes with SPP1⁺ macrophages, reinforces desmoplasia and T cell dysfunction, and predicts poor prognosis.¹⁰¹ In HCC, the liver template is shaped by fibrotic parenchyma and stellate-cell/portal-fibroblast reservoirs.^{26,29,75} Spatial analyses identify distinct marginal and intratumoral stromal architectures with divergent outcomes⁸³; SPP1-high cancer cells near stellate cells can promote stellate-to-CAF conversion through CD44-PI3K/AKT signaling,¹⁰² and CAFs within and around cancer nests support cancer-cell stemness.¹⁰³ Thus, archetype G provides the tissue-specific coordinate system for interpreting other archetypes, emphasizing that CAF biomarkers and targeting strategies should be evaluated in relation to lineage reservoir, anatomical compartment, and organ-specific vulnerability rather than generic CAF abundance alone.

SPATIOTEMPORAL EVOLUTION OF CAF SPATIAL ARCHETYPES

CAF spatial archetypes are not fixed cellular assemblies but dynamic configurations that shift over space and time during tumor

progression and under therapeutic stress.^{9,10} Their emergence, persistence, and transition are shaped by local tissue architecture, neighboring-cell composition, and changing signaling environments. However, our understanding of CAF temporal plasticity in human tumors remains limited by the lack of longitudinal sampling; evidence is often inferred from cross-sectional spatial datasets, *in silico* trajectory analyses, or comparisons between treatment-naïve and post-therapy cohorts.^{8,104} Despite these limitations, available data suggest that stromal responses evolve during progression and that distinct archetypes often coexist as coupled modules. For example, breast tumors contain distal iCAF-macrophage niches alongside cancer-proximal myCAF territories^{8,19,104}; human lung tumors show stage-associated boundary CAF layers and matrix-remodeling CAFs linked to T cell exclusion²⁰; and HCC contains spatially distinct intratumoral inflammatory CAF hubs and marginal wound-healing stromal compartments with divergent clinical associations.⁸³ These observations support a model in which early lesions may contain reactive fibroblast programs with wound-healing, inflammatory, or partial-containment features, whereas advanced lesions more frequently exhibit archetype A at the cancer-stroma border and archetype E at the invasive front, with more established immune-exclusion modules as stromal barriers mature. Thus, temporal evolution in the stroma may be reflected not only in changes in CAF state but also in shifts in the relative prominence, coupling, and plasticity of spatial archetypes. These shifts likely reflect changing balances among ecosystem cues, including TGF- β , inflammatory chemokines/cytokines, interferons, mechanical stress, hypoxia, therapy-induced damage, and the states of surrounding cancer and immune cells. Different archetypes may therefore vary in their degree of plasticity: inflammatory and immune-adjacent archetypes, such as archetypes B and D, may be rapidly induced by cytokine, interferon, or therapy-associated cues, whereas ECM-rich archetypes A and E may become more structurally persistent once matrix architecture is established. Under therapeutic pressure, such transitions can be accelerated or redirected, reshaping TME spatial organization, influencing therapy response and patient stratification, as discussed in the following sections.

TRANSLATING PHENOTYPE AND SPATIAL PERSPECTIVES TO CLINICAL STRATEGY

CAF-enriched niches undergo spatial and functional reorganization in treated tumors. Cytotoxic therapy, radiation, and immunotherapy can redirect CAFs toward barrier-forming, inflammatory, senescent, or immune-supportive states, with consequences that differ across tissue context and treatment modality. In rectal cancer, chemoradiation induces IL-1 α -dependent iCAF polarization and senescence associated with resistance, whereas neoadjuvant chemotherapy or radiotherapy can remodel CAF subsets toward immune-recruiting programs linked to improved response.^{46,105,106} In high-grade serous ovarian cancer, residual ANTXR1⁺ myCAFs after chemotherapy suppress CD8⁺ T cell cytotoxicity through YAP1 signaling and correlate with poor prognosis.¹³ In PDAC, spatial profiling of neoadjuvant-treated tumors reveals therapy-associated remodeling of CAF-cancer-cell interactions, including IL-6 family signaling linked to

chemoresistance.¹⁰⁷ In NSCLC, neoadjuvant chemoimmunotherapy can expand IFN- γ -induced apCAF-like states in fibrotic hotspots, where they are associated with regulatory T cell programs and impaired response.⁶¹ Similar context-dependent remodeling has been observed in other treated cohorts: in head and neck cancer, paired pre/post nivolumab samples identified immunostimulatory CAF subsets associated with response¹⁰⁸; in esophageal cancer, neoadjuvant anti-PD-1 plus chemotherapy expanded IL-6⁺CCL2⁺ and CD248⁺ CAF states linked to resistance¹⁰⁹; and in bladder cancer, chemotherapy-induced PTGER3⁺ CAFs enhanced CD8⁺ T cell cytotoxicity and correlated with favorable response.¹¹⁰ Together, these studies indicate that treatment reshapes archetype composition, producing resistance-associated, response-associated, or compensatory niches. CAF-targeted strategies therefore need to pair baseline archetype detection with post-treatment spatial monitoring to assess niche persistence, compensation, and response prediction.

Preclinical and clinical approaches to stroma-targeted therapy

Despite strong evidence linking CAFs to immunosuppression and therapy resistance, stroma-directed therapies have been difficult to translate into clinical benefit. Most preclinical strategies fall into four categories: direct depletion, reprogramming, functional inhibition, and ECM targeting (Figure 4). Importantly, only a small number have reached clinical testing, and among those, none has demonstrated unambiguous benefit. Understanding why requires examining the therapeutic hypothesis of each program and whether the drug achieved its intended pharmacodynamic effect.

Direct depletion has largely focused on FAP, but FAP lacks the specificity required for selective CAF targeting: it is expressed across multiple CAF subsets and is also present on FRCs in lymph nodes, bone marrow, and skeletal muscle stromal cells. As a result, systemic depletion of FAP⁺ cells in mice disrupted lymphoid architecture, impaired immune cell homing, caused cachexia and anemia.^{111,112} Clinically, sibrotuzumab (unconjugated anti-FAP) showed tumor-specific uptake but no objective responses in a phase II trial, as the unconjugated antibody lacked a cytotoxic mechanism.^{113,114} FAP-targeted radioligands pursue a different hypothesis: concentrating radiation in the TME via the crossfire effect rather than killing radio-resistant CAFs, with early phase I safety and modest disease-control signals.¹¹⁵ More selective targets may offer a better path: LRRC15 marks a TGF- β -driven myCAF subset absent from normal tissue, and genetic depletion of LRRC15⁺ CAFs recalibrated the stroma toward immune-permissive states in mice.^{7,25} However, the only LRRC15-targeting agent tested clinically, the antibody-drug conjugate (ADC) ABBV-085, was designed to target LRRC15⁺ cancer cells in sarcoma.¹¹⁶ LRRC15 remains a compelling CAF-directed target, but achieving this clinically may require payloads or modalities capable of eliminating CAFs.

Reprogramming strategies aim to restore quiescence. Vitamin D receptor agonism showed strong preclinical rationale,¹¹⁷ but multiple phase 2 trials of paricalcitol in PDAC have been negative, maybe due to insufficient on-target activation. All-trans retinoic acid is more encouraging: the STARPAC phase Ib trial es-

tablished a safe dose with MRI evidence of stromal modulation, and phase 2 (STARPAC2) is ongoing.¹¹⁸ Nonetheless, CAF heterogeneity makes outcomes of these reprogramming approaches difficult to predict without considering lineage and spatial context.

Functional inhibition has produced mixed clinical results. Broad TGF- β blockade has underperformed (bintrafusp alfa failed in phase 3 NSCLC with increased toxicity¹¹⁹), though isoform-selective approaches such as SRK-181 (anti-latent TGF- β 1), which eliminated cardiac toxicity in preclinical models,¹²⁰ are now in clinical testing with early signals of activity in anti-PD-1-resistant tumors (NCT04291079). CXCL12-CXCR4 blockade increased CD8⁺ T cell infiltration and showed a 32% objective response rate (ORR) in PDAC (COMBAT trial, motixafortide + pembrolizumab + chemotherapy), though randomized confirmation is lacking.¹²¹ IL-6 targeting has consistently disappointed, potentially reflecting cytokine redundancy in the inflammatory TME.^{122,123}

ECM-targeting strategies have produced the field's most instructive lessons. Hedgehog inhibition in PDAC illustrates a preclinical-clinical disconnect: preclinically, *Shh* deletion accelerated tumors in mice, establishing that stroma can restrain progression.⁶ Clinically, however, Hedgehog pathway inhibitor vismodegib showed no effect on stroma. Paired biopsies showed no changes in α -SMA, E-cadherin, or CD45⁺ markers, with outcomes identical to chemotherapy alone.^{124,125} PEGPH20 that degrades hyaluronic acid (HA) presents a different lesson: the drug confirmed target engagement (34%–80% tumor HA depletion, increased perfusion on imaging) and showed a higher ORR, but the phase 3 HALO-301 trial (HA-high selected) showed identical overall survival with shorter duration of response in the treatment arm,^{126,127} demonstrating that achieving the intended pharmacodynamic effect does not guarantee benefit when the therapeutic hypothesis is incomplete. Simtuzumab (anti-LOXL2) failed across six indications, but subsequent studies revealed the antibody likely did not inhibit LOXL2 enzymatic activity,¹²⁸ making this a drug-level failure, though whether LOXL2 itself is a viable therapeutic target in this context remains unresolved.

Spatial archetype-guided CAF targeting

The shift toward spatial archetypes reframes CAFs as components of recurring multicellular configurations rather than isolated cell states. Translating this into therapy requires addressing three priorities.

A first priority is to define how CAF spatial archetypes emerge and rewire during tumor progression and under therapy. Although recurring archetypes are becoming clearer, their temporal logic remains poorly understood. This process may begin before overt malignancy, as fibroblast remodeling in precancerous lesions can establish conditions that support epithelial survival and progression.² During therapy, archetypes may regress, persist, or be replaced by compensatory stromal configurations.^{55,61,62,80,105–107,129} These observations suggest that archetype plasticity is shaped by tissue context, treatment modality, and CAF lineage, and that the same broad archetype may be response-associated or resistance-associated. Defining when and how these stromal configurations shift will require longitudinal, site-matched sampling across the disease continuum,

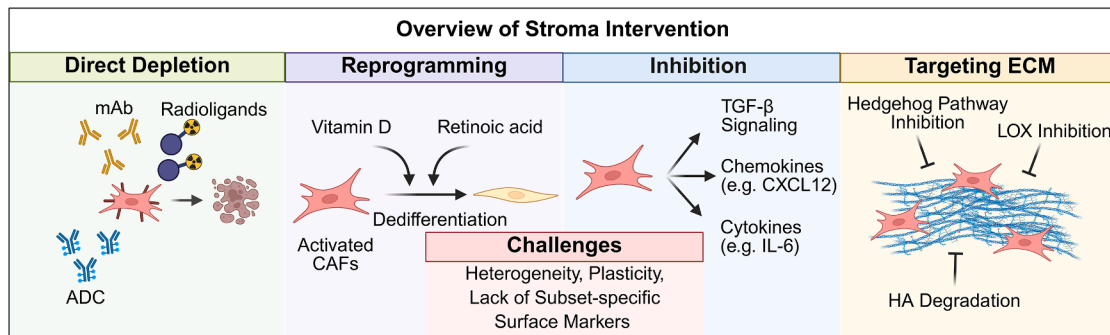


Figure 4. Overview of stroma-targeted therapies

Schematic summarizing four categories of current stroma-targeted intervention strategies and their associated challenges. Direct depletion (left) includes monoclonal antibodies (mAb), radioligands, and antibody-drug conjugates (ADC) targeting CAF surface markers such as FAP. Reprogramming (center-left) employs agents such as vitamin D and retinoic acid to induce dedifferentiation of activated CAFs toward a quiescent state. Inhibition (center-right) targets key CAF-derived signaling pathways, including TGF- β signaling, chemokines (e.g., CXCL12), and cytokines (e.g., IL-6). ECM targeting (right) aims to disrupt the tumor-promoting extracellular matrix through Hedgehog pathway inhibition, LOX inhibition, and hyaluronan (HA) degradation. Central box highlights the key challenges limiting these approaches: stromal heterogeneity, phenotypic plasticity, and lack of subset-specific surface markers that complicate selective therapeutic targeting.

particularly in precancerous, neoadjuvant, and metastatic settings, with study designs that capture reciprocity between CAFs and neighboring tumor, immune, and vascular compartments.

A second priority is the development of niche-targeted interventions that suppress tumor-supportive stromal functions without disrupting CAF states that contribute to immune support or treatment responsiveness, with awareness that the fibroblast compartment will adapt to intervention. Emerging preclinical targets suggest that niche-specific modulation may be achievable through multiple molecular entry points. *ADAM12* ablation in CAFs elicits IFN-I-responsive programs and induces tumor rejection in mice.¹³⁰ An orally bioavailable NNMT inhibitor restored checkpoint blockade efficacy through stromal-specific metabolic reprogramming.¹³¹ And 11-HETE has been linked to niche-specific lipid signaling in the CAF compartment.¹¹⁰ Among clinically testable approaches, selective TGF- β inhibition or LRRC15 ADC combined with ICB in archetype A-enriched tumors may represent the more immediate translational paths.

A third priority is the development of standardized spatial metrics and practical detection strategies for clinical translation. CAF archetypes should not be inferred from single biomarkers alone but from the combined assessment of CAF state, anatomical position, neighboring cell composition, and ECM architecture. In practice, multiplex IHC, imaging mass cytometry, or spatial proteomics could quantify CAF-cancer cell, CAF-immune cell, CAF-vessel, and CAF-ECM relationships using CAF-state markers together with epithelial, endothelial, mural, immune, and matrix markers. Spatial transcriptomics can define archetype-associated gene programs, ligand-receptor neighborhoods, and regional gradients, whereas scRNA-seq or cellular indexing of transcriptomes and epitopes by sequencing (CITE-seq) can resolve the CAF states and surface markers that populate each spatial compartment. Applying these readouts to paired baseline and post-treatment samples would allow investigators to assess whether an archetype regresses, persists, or is replaced by a compensatory niche. Integrating these measurements with computational pathology and clinical imaging

may generate spatial risk models that improve patient stratification, guide stromal-targeted combinations, and support more precise monitoring of treatment response. Ultimately, the goal is not simply to describe stromal complexity, but to modulate it in a clinically meaningful way, moving from static annotation to mechanism-informed intervention in which stromal niches are selectively disrupted, preserved, or reprogrammed to support durable tumor control.

CAF-related biomarkers for diagnosis and monitoring

The distinct molecular programs and spatial architectures of CAFs present promising biomarker opportunities for assessing disease progression and therapeutic response. While FAP-based PET/CT imaging demonstrates diagnostic utility in delineating tumor mass from normal tissue,^{132,133} emerging markers like ANTXR1 hold promise for next-generation, high-resolution PET imaging and precision radioligand therapy.¹³⁴ Complementing these targeted molecular approaches, advances in computational pathology and radiomics now facilitate the translation of complex CAF spatial archetypes into accessible clinical risk-stratification tools, utilizing standard H&E or medical imaging to guide diagnosis, prognosis, and treatment planning.^{10,135} In parallel, circulating CAF-related biomarkers, such as cfDNA, cfRNA, and extracellular vesicles, have emerged as a minimally invasive strategy to monitor CAF dynamics.^{136–138} However, the routine clinical utility of these circulating readouts will heavily depend on rigorous validation across diverse tumor types and treatment settings, alongside the strict standardization of detection methods, analytical pipelines, and interpretation frameworks.

CONCLUSION AND FUTURE OUTLOOK

In this review, we propose a multi-layered view of CAF biology organized around conserved molecular phenotypes, a hierarchical classification system, and spatial archetypes that capture recurring patterns of CAF localization and microenvironmental association across solid tumors. A central theme is that CAF identity is the product of lineage origin, signaling context, and

spatial positioning within the TME, which helps explain why strategies targeting generic markers or single pathways have largely failed and argues for spatially informed therapeutic approaches.

Key unresolved questions include the precise origins of certain CAF subtypes, the context-dependent function of distinct CAF programs, the temporal dynamics of spatial archetypes across the disease continuum, and the translation of spatial descriptors into standardized clinical biomarkers. A particularly urgent gap is the lack of highly specific surface markers that uniquely identify pro-disease CAF subsets. Addressing these gaps will require a broader experimental toolkit, including lineage tracing, longitudinal site-matched sampling, perturbation studies, surface proteomics, spatial omics, computational pathology, and biomarker-driven clinical trial design.

The convergence of pan-cancer atlases, high-resolution spatial technologies, and emerging niche-targeted agents creates an opportunity to move from describing stromal complexity to modulating it. Selectively targeting CAF programs within their native spatial niches may ultimately be required to achieve durable tumor control.

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H.H. collected initial input from all authors. L.X., Z.Z., and L.T. organized the literature. Y.L., X.C., Y.D., Y.J., G.B., M.H.S., L.W., and H.H. wrote the manuscript. All authors contributed ideas and assisted with editing and reviewing the article. Y.L., X.C., and Y.D. contributed equally.

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DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

The authors used ChatGPT-5 to improve conciseness and enhance clarity. After using this tool, the authors reviewed and edited the content as needed. The authors take full responsibility for the integrity of generated content.

REFERENCES

- Dvorak, H.F. (1986). Tumors: wounds that do not heal. Similarities between tumor stroma generation and wound healing. *N. Engl. J. Med.* 315, 1650–1659. <https://doi.org/10.1056/NEJM198612253152606>.
- Sahai, E., Astsaturov, I., Cukierman, E., DeNardo, D.G., Egeblad, M., Evans, R.M., Fearon, D., Gretchen, F.R., Hingorani, S.R., Hunter, T., et al. (2020). A framework for advancing our understanding of cancer-associated fibroblasts. *Nat. Rev. Cancer* 20, 174–186. <https://doi.org/10.1038/s41568-019-0238-1>.
- Biffi, G., and Tuveson, D.A. (2021). Diversity and Biology of Cancer-Associated Fibroblasts. *Physiol. Rev.* 101, 147–176. <https://doi.org/10.1152/physrev.00048.2019>.
- Cheng, P.S.W., Zaccaria, M., and Biffi, G. (2025). Functional heterogeneity of fibroblasts in primary tumors and metastases. *Trends Cancer* 11, 135–153. <https://doi.org/10.1016/j.trecan.2024.11.005>.
- Özdemir, B., Pentcheva-Hoang, T., Carstens, J.L., Zheng, X., Wu, C.C., Simpson, T.R., Laklai, H., Sugimoto, H., Kahlert, C., Novitskiy, S.V., et al. (2014). Depletion of carcinoma-associated fibroblasts and fibrosis induces immunosuppression and accelerates pancreas cancer with reduced survival. *Cancer Cell* 25, 719–734. <https://doi.org/10.1016/j.ccr.2014.04.005>.
- Rhim, A.D., Oberstein, P.E., Thomas, D.H., Mirek, E.T., Palermo, C.F., Sastra, S.A., Dekleva, E.N., Saunders, T., Becerra, C.P., Tattersall, I. W., et al. (2014). Stromal elements act to restrain, rather than support, pancreatic ductal adenocarcinoma. *Cancer Cell* 25, 735–747. <https://doi.org/10.1016/j.ccr.2014.04.021>.
- Buechler, M.B., Pradhan, R.N., Krishnamurthy, A.T., Cox, C., Calviello, A. K., Wang, A.W., Yang, Y.A., Tam, L., Caothien, R., Roose-Girma, M., et al. (2021). Cross-tissue organization of the fibroblast lineage. *Nature* 593, 575–579. <https://doi.org/10.1038/s41586-021-03549-5>.
- Croizer, H., Mhaidly, R., Kieffer, Y., Gentric, G., Djerroudi, L., Leclerc, R., Pelon, F., Robley, C., Bohec, M., Meng, A., et al. (2024). Deciphering the spatial landscape and plasticity of immunosuppressive fibroblasts in breast cancer. *Nat. Commun.* 15, 2806. <https://doi.org/10.1038/s41467-024-47068-z>.
- Friedman, G., Levi-Galibov, O., David, E., Bornstein, C., Giladi, A., Dadiani, M., Mayo, A., Halperin, C., Pevsner-Fischer, M., Lavon, H., et al. (2020). Cancer-associated fibroblast compositions change with breast cancer progression linking the ratio of S100A4(+) and PDPN(+) CAFs to clinical outcome. *Nat. Cancer* 1, 692–708. <https://doi.org/10.1038/s43018-020-0082-y>.
- Liu, Y., Sinjab, A., Min, J., Han, G., Paradiso, F., Zhang, Y., Wang, R., Pei, G., Dai, Y., Liu, Y., et al. (2025). Conserved spatial subtypes and cellular neighborhoods of cancer-associated fibroblasts revealed by single-cell spatial multi-omics. *Cancer Cell* 43, 905–924.e6. <https://doi.org/10.1016/j.ccell.2025.03.004>.
- Chen, X., Zhou, Z., Xie, L., Qiao, K., Jia, Y., Liu, S., Yazgan, Z., Rossi, F., Liu, Y., Zhang, B., et al. (2025). Single-cell resolution spatial analysis of antigen-presenting cancer-associated fibroblast niches. *Cancer Cell* 43, 2224–2240.e9. <https://doi.org/10.1016/j.ccell.2025.09.001>.
- Dominguez, C.X., Müller, S., Keerthivasan, S., Koeppen, H., Hung, J., Gierke, S., Breart, B., Foreman, O., Bainbridge, T.W., Castiglioni, A., et al. (2020). Single-Cell RNA Sequencing Reveals Stromal Evolution into LRRC15(+) Myofibroblasts as a Determinant of Patient Response to Cancer Immunotherapy. *Cancer Discov.* 10, 232–253. <https://doi.org/10.1158/2159-8290.CD-19-0644>.
- Licaj, M., Mhaidly, R., Kieffer, Y., Croizer, H., Bonneau, C., Meng, A., Djerroudi, L., Mujangi-Ebeka, K., Hocine, H.R., Bourachot, B., et al. (2024). Residual ANTXR1+ myofibroblasts after chemotherapy inhibit anti-tumor immunity via YAP1 signaling pathway. *Nat. Commun.* 15, 1312. <https://doi.org/10.1038/s41467-024-45595-3>.
- Pei, G., Min, J., Rajapakshe, K.I., Branchi, V., Liu, Y., Selvaranesan, B.C., Thege, F., Sadeghian, D., Zhang, D., Cho, K.S., et al. (2025). Spatial mapping of transcriptomic plasticity in metastatic pancreatic cancer. *Nature* 642, 212–221. <https://doi.org/10.1038/s41586-025-08927-x>.
- Biffi, G., Oni, T.E., Spielman, B., Hao, Y., Elyada, E., Park, Y., Preall, J., and Tuveson, D.A. (2019). IL1-Induced JAK/STAT Signaling Is Antagonized by TGFβ to Shape CAF Heterogeneity in Pancreatic Ductal Adenocarcinoma. *Cancer Discov.* 9, 282–301. <https://doi.org/10.1158/2159-8290.CD-18-0710>.
- Öhlund, D., Handly-Santana, A., Biffi, G., Elyada, E., Almeida, A.S., Ponz-Sarvisé, M., Corbo, V., Oni, T.E., Hearn, S.A., Lee, E.J., et al. (2017).

Distinct populations of inflammatory fibroblasts and myofibroblasts in pancreatic cancer. *J. Exp. Med.* 214, 579–596. <https://doi.org/10.1084/jem.20162024>.

17. Zhang, Y., Lazarus, J., Steele, N.G., Yan, W., Lee, H.J., Nwosu, Z.C., Halbrook, C.J., Menjivar, R.E., Kemp, S.B., Sirihorachai, V.R., et al. (2020). Regulatory T-cell Depletion Alters the Tumor Microenvironment and Accelerates Pancreatic Carcinogenesis. *Cancer Discov.* 10, 422–439. <https://doi.org/10.1158/2159-8290.CD-19-0958>.
18. Costa, A., Kieffer, Y., Scholer-Dahirel, A., Pelon, F., Bourachot, B., Cardon, M., Sirven, P., Magagna, I., Fuhrmann, L., Bernard, C., et al. (2018). Fibroblast Heterogeneity and Immunosuppressive Environment in Human Breast Cancer. *Cancer Cell* 33, 463–479.e10. <https://doi.org/10.1016/j.ccell.2018.01.011>.
19. Kieffer, Y., Hocine, H.R., Gentric, G., Pelon, F., Bernard, C., Bourachot, B., Lameiras, S., Albergant, L., Bonneau, C., Guyard, A., et al. (2020). Single-Cell Analysis Reveals Fibroblast Clusters Linked to Immunotherapy Resistance in Cancer. *Cancer Discov.* 10, 1330–1351. <https://doi.org/10.1158/2159-8290.CD-19-1384>.
20. Grout, J.A., Sirven, P., Leader, A.M., Maskey, S., Hector, E., Puisieux, I., Steffan, F., Cheng, E., Tung, N., Maurin, M., et al. (2022). Spatial Positioning and Matrix Programs of Cancer-Associated Fibroblasts Promote T-cell Exclusion in Human Lung Tumors. *Cancer Discov.* 12, 2606–2625. <https://doi.org/10.1158/2159-8290.CD-21-1714>.
21. Shani, O., Raz, Y., Monteran, L., Scharff, Y., Levi-Galibov, O., Megides, O., Shacham, H., Cohen, N., Silverbush, D., Avivi, C., et al. (2021). Evolution of fibroblasts in the lung metastatic microenvironment is driven by stage-specific transcriptional plasticity. *eLife* 10, e60745. <https://doi.org/10.7554/eLife.60745>.
22. Shaashua, L., Ben-Shmuel, A., Pevsner-Fischer, M., Friedman, G., Levi-Galibov, O., Nandakumar, S., Barki, D., Nevo, R., Brown, L.E., Zhang, W., et al. (2022). BRCA mutational status shapes the stromal microenvironment of pancreatic cancer linking clusterin expression in cancer associated fibroblasts with HSF1 signaling. *Nat. Commun.* 13, 6513. <https://doi.org/10.1038/s41467-022-34081-3>.
23. Arina, A., Idel, C., Hyjek, E.M., Alegre, M.L., Wang, Y., Bindokas, V.P., Weichselbaum, R.R., and Schreiber, H. (2016). Tumor-associated fibroblasts predominantly come from local and not circulating precursors. *Proc. Natl. Acad. Sci. USA* 113, 7551–7556. <https://doi.org/10.1073/pnas.1600363113>.
24. Raz, Y., Cohen, N., Shani, O., Bell, R.E., Novitskiy, S.V., Abramovitz, L., Levy, C., Milyavsky, M., Leider-Trejo, L., Moses, H.L., et al. (2018). Bone marrow-derived fibroblasts are a functionally distinct stromal cell population in breast cancer. *J. Exp. Med.* 215, 3075–3093. <https://doi.org/10.1084/jem.20180818>.
25. Krishnamurthy, A.T., Shyer, J.A., Thai, M., Gandham, V., Buechler, M.B., Yang, Y.A., Pradhan, R.N., Wang, A.W., Sanchez, P.L., Qu, Y., et al. (2022). LRRc15(+) myofibroblasts dictate the stromal setpoint to suppress tumour immunity. *Nature* 611, 148–154. <https://doi.org/10.1038/s41586-022-05272-1>.
26. Wang, S.S., Yuan, J., Tang, X.T., Yin, X., Fang, K., Chen, L.V., Ren, Z., and Zhou, B.O. (2025). Periductal fibroblasts participate in liver homeostasis, fibrosis, and tumorigenesis. *J. Exp. Med.* 222, e20232123. <https://doi.org/10.1084/jem.20232123>.
27. Garcia, P.E., Adoumie, M., Kim, E.C., Zhang, Y., Scales, M.K., El-Tawil, Y.S., Shaikh, A.Z., Wen, H.J., Bednar, F., Allen, B.L., et al. (2020). Differential Contribution of Pancreatic Fibroblast Subsets to the Pancreatic Cancer Stroma. *Cell. Mol. Gastroenterol. Hepatol.* 10, 581–599. <https://doi.org/10.1016/j.jcmgh.2020.05.004>.
28. Affo, S., Nair, A., Brundu, F., Ravichandra, A., Bhattacharjee, S., Matsuda, M., Chin, L., Filliol, A., Wen, W., Song, X., et al. (2021). Promotion of cholangiocarcinoma growth by diverse cancer-associated fibroblast subpopulations. *Cancer Cell* 39, 866–882.e11. <https://doi.org/10.1016/j.ccell.2021.03.012>.
29. Filliol, A., Saito, Y., Nair, A., Dapito, D.H., Yu, L.X., Ravichandra, A., Bhattacharjee, S., Affo, S., Fujiwara, N., Su, H., et al. (2022). Opposing roles of hepatic stellate cell subpopulations in hepatocarcinogenesis. *Nature* 610, 356–365. <https://doi.org/10.1038/s41586-022-05289-6>.
30. Helms, E.J., Berry, M.W., Chaw, R.C., DuFort, C.C., Sun, D., Onate, M.K., Oon, C., Bhattacharyya, S., Sanford-Crane, H., Horton, W., et al. (2022). Mesenchymal Lineage Heterogeneity Underlies Nonredundant Functions of Pancreatic Cancer-Associated Fibroblasts. *Cancer Discov.* 12, 484–501. <https://doi.org/10.1158/2159-8290.CD-21-0601>.
31. Huang, H., Wang, Z., Zhang, Y., Pradhan, R.N., Ganguly, D., Chandra, R., Murimwa, G., Wright, S., Gu, X., Maddipati, R., et al. (2022). Mesothelial cell-derived antigen-presenting cancer-associated fibroblasts induce expansion of regulatory T cells in pancreatic cancer. *Cancer Cell* 40, 656–673.e7. <https://doi.org/10.1016/j.ccell.2022.04.011>.
32. Hosaka, K., Yang, Y., Seki, T., Fischer, C., Dubey, O., Fredlund, E., Hartman, J., Religa, P., Morikawa, H., Ishii, Y., et al. (2016). Pericyte-fibroblast transition promotes tumor growth and metastasis. *Proc. Natl. Acad. Sci. USA* 113, E5618–E5627. <https://doi.org/10.1073/pnas.1608384113>.
33. Onder, L., Papadopoulos, C., Lütge, A., Cheng, H.W., Lütge, M., Perez-Shibayama, C., Gil-Cruz, C., De Martin, A., Kurz, L., Cadosch, N., et al. (2025). Fibroblastic reticular cells generate protective intratumoral T cell environments in lung cancer. *Cell* 188, 430–446.e20. <https://doi.org/10.1016/j.cell.2024.10.042>.
34. Du, Y., Shi, J., Wang, J., Xun, Z., Yu, Z., Sun, H., Bao, R., Zheng, J., Li, Z., and Ye, Y. (2024). Integration of Pan-Cancer Single-Cell and Spatial Transcriptomics Reveals Stromal Cell Features and Therapeutic Targets in Tumor Microenvironment. *Cancer Res.* 84, 192–210. <https://doi.org/10.1158/0008-5472.CAN-23-1418>.
35. Lior, C., Barki, D., Halperin, C., Iacobuzio-Donahue, C.A., Kelsen, D., and Shouval, R.S. (2024). Mapping the tumor stress network reveals dynamic shifts in the stromal oxidative stress response. *Cell Rep.* 43, 114236. <https://doi.org/10.1016/j.celrep.2024.114236>.
36. Dejana, E., and Lampugnani, M.G. (2018). Endothelial cell transitions. *Science* 362, 746–747. <https://doi.org/10.1126/science.aas9432>.
37. Sun, X., Cai, W., Li, H., Gao, C., Ma, X., Guo, Y., Fu, D., Xiao, D., Zhang, Z., Wang, Y., et al. (2025). Endothelial-like cancer-associated fibroblasts facilitate pancreatic cancer metastasis via vasculogenic mimicry and paracrine signalling. *Gut* 74, 1437–1451. <https://doi.org/10.1136/gutjnl-2024-333638>.
38. Hutton, C., Heider, F., Blanco-Gomez, A., Banyard, A., Kononov, A., Zhang, X., Karim, S., Paulus-Hock, V., Watt, D., Steele, N., et al. (2021). Single-cell analysis defines a pancreatic fibroblast lineage that supports anti-tumor immunity. *Cancer Cell* 39, 1227–1244.e20. <https://doi.org/10.1016/j.ccell.2021.06.017>.
39. Elyada, E., Bolisetty, M., Laise, P., Flynn, W.F., Courtois, E.T., Burkhart, R.A., Teinor, J.A., Belleau, P., Biffi, G., Lucito, M.S., et al. (2019). Cross-Species Single-Cell Analysis of Pancreatic Ductal Adenocarcinoma Reveals Antigen-Presenting Cancer-Associated Fibroblasts. *Cancer Discov.* 9, 1102–1123. <https://doi.org/10.1158/2159-8290.CD-19-0094>.
40. Mucciolo, G., Curcio, C., Roux, C., Li, W.Y., Capello, M., Curto, R., Chiariello, R., Giordano, D., Satolli, M.A., Lawlor, R., et al. (2021). IL17A critically shapes the transcriptional program of fibroblasts in pancreatic cancer and switches on their protumorigenic functions. *Proc. Natl. Acad. Sci. USA* 118, e2020395118. <https://doi.org/10.1073/pnas.2020395118>.
41. Bianchi, A., De Castro Silva, I., Deshpande, N.U., Singh, S., Mehra, S., Garrido, V.T., Guo, X., Nivelio, L.A., Kolonias, D.S., Saigh, S.J., et al. (2023). Cell-Autonomous Cxcl1 Sustains Tolerogenic Circuitries and Stromal Inflammation via Neutrophil-Derived TNF in Pancreatic Cancer. *Cancer Discov.* 13, 1428–1453. <https://doi.org/10.1158/2159-8290.CD-22-1046>.
42. Picard, F.S.R., Lutz, V., Brichkina, A., Neuhaus, F., Ruckebrod, T., Hupfer, A., Raifer, H., Klein, M., Bopp, T., Pfefferle, P.I., et al. (2023). IL-17A-producing CD8(+) T cells promote PDAC via induction of inflammatory cancer-associated fibroblasts. *Gut* 72, 1510–1522. <https://doi.org/10.1136/gutjnl-2022-327855>.
43. Lee, B.Y., Hogg, E.K.J., Below, C.R., Kononov, A., Blanco-Gomez, A., Heider, F., Xu, J., Hutton, C., Zhang, X., Scheidt, T., et al. (2021). Heterocellular OSM-OSMR signalling reprograms fibroblasts to promote pancreatic cancer growth and metastasis. *Nat. Commun.* 12, 7336. <https://doi.org/10.1038/s41467-021-27607-8>.
44. Schwörer, S., Cimino, F.V., Ros, M., Tsanov, K.M., Ng, C., Lowe, S.W., Carmona-Fontaine, C., and Thompson, C.B. (2023). Hypoxia Potentiates

- the Inflammatory Fibroblast Phenotype Promoted by Pancreatic Cancer Cell-Derived Cytokines. *Cancer Res.* 83, 1596–1610. <https://doi.org/10.1158/0008-5472.CAN-22-2316>.
45. Zhang, Y., Ling, L., Murad, R., Maganti, S., Manceau, A., Hetrick, H.A., Neff, M., Galapate, C.M., Grenier, S.F., Carrette, F., et al. (2025). Macropinocytosis maintains CAF subtype identity under metabolic stress in pancreatic cancer. *Cancer Cell* 43, 1677–1696.e15. <https://doi.org/10.1016/j.ccell.2025.06.021>.
46. Nicolas, A.M., Pesic, M., Engel, E., Ziegler, P.K., Diefenhardt, M., Kennel, K.B., Buettner, F., Conche, C., Petrocelli, V., Elwakeel, E., et al. (2022). Inflammatory fibroblasts mediate resistance to neoadjuvant therapy in rectal cancer. *Cancer Cell* 40, 168–184.e13. <https://doi.org/10.1016/j.ccell.2022.01.004>.
47. Su, S., Chen, J., Yao, H., Liu, J., Yu, S., Lao, L., Wang, M., Luo, M., Xing, Y., Chen, F., et al. (2018). CD10(+)GPR77(+) Cancer-Associated Fibroblasts Promote Cancer Formation and Chemoresistance by Sustaining Cancer Stemness. *Cell* 172, 841–856.e16. <https://doi.org/10.1016/j.cell.2018.01.009>.
48. Gao, Y., Li, J., Cheng, W., Diao, T., Liu, H., Bo, Y., Liu, C., Zhou, W., Chen, M., Zhang, Y., et al. (2024). Cross-tissue human fibroblast atlas reveals myofibroblast subtypes with distinct roles in immune modulation. *Cancer Cell* 42, 1764–1783.e10. <https://doi.org/10.1016/j.ccell.2024.08.020>.
49. Steele, N.G., Biffi, G., Kemp, S.B., Zhang, Y., Drouillard, D., Syu, L., Hao, Y., Oni, T.E., Brosnan, E., Elyada, E., et al. (2021). Inhibition of Hedgehog Signaling Alters Fibroblast Composition in Pancreatic Cancer. *Clin. Cancer Res.* 27, 2023–2037. <https://doi.org/10.1158/1078-0432.CCR-20-3715>.
50. Calvo, F., Ege, N., Grande-Garcia, A., Hooper, S., Jenkins, R.P., Chaudhry, S.I., Harrington, K., Williamson, P., Moeendarbary, E., Charas, G., and Sahai, E. (2013). Mechanotransduction and YAP-dependent matrix remodelling is required for the generation and maintenance of cancer-associated fibroblasts. *Nat. Cell Biol.* 15, 637–646. <https://doi.org/10.1038/ncb2756>.
51. Muhl, L., Genov  , G., Leptidis, S., Liu, J., He, L., Mocci, G., Sun, Y., Gustafsson, S., Buyandelger, B., Chivukula, I.V., et al. (2020). Single-cell analysis uncovers fibroblast heterogeneity and criteria for fibroblast and mural cell identification and discrimination. *Nat. Commun.* 11, 3953. <https://doi.org/10.1038/s41467-020-17740-1>.
52. Bhattacharjee, S., Hamberger, F., Ravichandra, A., Miller, M., Nair, A., Affo, S., Filliol, A., Chin, L., Savage, T.M., Yin, D., et al. (2021). Tumor restriction by type I collagen opposes tumor-promoting effects of cancer-associated fibroblasts. *J. Clin. Investig.* 131, e146987. <https://doi.org/10.1172/JCI146987>.
53. Mucciolo, G., Araos Henr  quez, J., Jihad, M., Pinto Teles, S., Manansala, J.S., Li, W., Ashworth, S., Lloyd, E.G., Cheng, P.S.W., Luo, W., et al. (2024). EGFR-activated myofibroblasts promote metastasis of pancreatic cancer. *Cancer Cell* 42, 101–118.e11. <https://doi.org/10.1016/j.ccell.2023.12.002>.
54. Raymant, M., Astuti, Y., Alvaro-Espinosa, L., Green, D., Quaranta, V., Bellomo, G., Glenn, M., Chandran-Gorner, V., Palmer, D.H., Halloran, C., et al. (2024). Macrophage-fibroblast JAK/STAT dependent crosstalk promotes liver metastatic outgrowth in pancreatic cancer. *Nat. Commun.* 15, 3593. <https://doi.org/10.1038/s41467-024-47949-3>.
55. Belle, J.I., Sen, D., Baer, J.M., Liu, X., Lander, V.E., Ye, J., Sells, B.E., Knolhoff, B.L., Faiz, A., Kang, L.I., et al. (2024). Senescence Defines a Distinct Subset of Myofibroblasts That Orchestrates Immunosuppression in Pancreatic Cancer. *Cancer Discov.* 14, 1324–1355. <https://doi.org/10.1158/2159-8290.CD-23-0428>.
56. Ye, J., Baer, J.M., Faget, D.V., Morikis, V.A., Ren, Q., Melam, A., Delgado, A.P., Luo, X., Bagchi, S.M., Belle, J.I., et al. (2024). Senescent CAFs Mediate Immunosuppression and Drive Breast Cancer Progression. *Cancer Discov.* 14, 1302–1323. <https://doi.org/10.1158/2159-8290.CD-23-0426>.
57. Hosein, A.N., Huang, H., Wang, Z., Parmar, K., Du, W., Huang, J., Maitra, A., Olson, E., Verma, U., and Brekken, R.A. (2019). Cellular heterogeneity during mouse pancreatic ductal adenocarcinoma progression at single-cell resolution. *JCI Insight* 4, e129212. <https://doi.org/10.1172/jci.insight.129212>.
58. Geppert, T.D., and Lipsky, P.E. (1985). Antigen presentation by interferon-gamma-treated endothelial cells and fibroblasts: differential ability to function as antigen-presenting cells despite comparable Ia expression. *J. Immunol.* 135, 3750–3762.
59. Mayer, L., and Shlien, R. (1987). Evidence for function of Ia molecules on gut epithelial cells in man. *J. Exp. Med.* 166, 1471–1483. <https://doi.org/10.1084/jem.166.5.1471>.
60. Steimle, V., Otten, L.A., Zufferey, M., and Mach, B. (1993). Complementation cloning of an MHC class II transactivator mutated in hereditary MHC class II deficiency (or bare lymphocyte syndrome). *Cell* 75, 135–146.
61. Cao, Z., Meng, Z., Li, J., Tian, Y., Lu, L., Wang, A., Huang, J., Wang, J., Sun, J., Chen, L., et al. (2025). Interferon-gamma-stimulated antigen-presenting cancer-associated fibroblasts hinder neoadjuvant chemioimmunotherapy efficacy in lung cancer. *Cell Rep. Med.* 6, 102017. <https://doi.org/10.1016/j.xcrm.2025.102017>.
62. Song, J., Wei, R., Liu, C., Zhao, Z., Liu, X., Wang, Y., Liu, F., and Liu, X. (2025). Antigen-presenting cancer associated fibroblasts enhance anti-tumor immunity and predict immunotherapy response. *Nat. Commun.* 16, 2175. <https://doi.org/10.1038/s41467-025-57465-7>.
63. Kerdidani, D., Aerakis, E., Verrou, K.M., Angelidis, I., Douka, K., Maniou, M.A., Stamoulis, P., Goudevenou, K., Prados, A., Tzaferis, C., et al. (2022). Lung tumor MHCII immunity depends on in situ antigen presentation by fibroblasts. *J. Exp. Med.* 219, e20210815. <https://doi.org/10.1084/jem.20210815>.
64. Saut  s-Fridman, C., Petitprez, F., Calderaro, J., and Fridman, W.H. (2019). Tertiary lymphoid structures in the era of cancer immunotherapy. *Nat. Rev. Cancer* 19, 307–325. <https://doi.org/10.1038/s41568-019-0144-6>.
65. Cords, L., Tietscher, S., Anzeneder, T., Langwieder, C., Rees, M., de Souza, N., and Bodenmiller, B. (2023). Cancer-associated fibroblast classification in single-cell and spatial proteomics data. *Nat. Commun.* 14, 4294. <https://doi.org/10.1038/s41467-023-3762-1>.
66. Foster, D.S., Januszyk, M., Delitto, D., Yost, K.E., Griffin, M., Guo, J., Guardino, N., Delitto, A.E., Chinta, M., Burcham, A.R., et al. (2022). Multitomic analysis reveals conservation of cancer-associated fibroblast phenotypes across species and tissue of origin. *Cancer Cell* 40, 1392–1406.e7. <https://doi.org/10.1016/j.ccell.2022.09.015>.
67. Wang, Y., Liang, Y., Xu, H., Zhang, X., Mao, T., Cui, J., Yao, J., Wang, Y., Jiao, F., Xiao, X., et al. (2021). Single-cell analysis of pancreatic ductal adenocarcinoma identifies a novel fibroblast subtype associated with poor prognosis but better immunotherapy response. *Cell Discov.* 7, 36. <https://doi.org/10.1038/s41421-021-00271-4>.
68. Xing, X., Yang, F., Huang, Q., Guo, H., Li, J., Qiu, M., Bai, F., and Wang, J. (2021). Decoding the multicellular ecosystem of lung adenocarcinoma manifested as pulmonary subsolid nodules by single-cell RNA sequencing. *Sci. Adv.* 7, eabd9738. <https://doi.org/10.1126/sciadv.abd9738>.
69. Wu, S.Z., Roden, D.L., Wang, C., Holliday, H., Harvey, K., Cazet, A.S., Murphy, K.J., Pereira, B., Al-Eryani, G., Bartonicek, N., et al. (2020). Stromal cell diversity associated with immune evasion in human triple-negative breast cancer. *EMBO J.* 39, e104063. <https://doi.org/10.15252/emboj.2019104063>.
70. Cords, L., de Souza, N., and Bodenmiller, B. (2024). Classifying cancer-associated fibroblasts-The good, the bad, and the target. *Cancer Cell* 42, 1480–1485. <https://doi.org/10.1016/j.ccell.2024.08.011>.
71. Croft, A.P., Campos, J., Jansen, K., Turner, J.D., Marshall, J., Attar, M., Savary, L., Wehmeyer, C., Naylor, A.J., Kemble, S., et al. (2019). Distinct fibroblast subsets drive inflammation and damage in arthritis. *Nature* 570, 246–251. <https://doi.org/10.1038/s41586-019-1263-7>.
72. Kinchen, J., Chen, H.H., Parikh, K., Antanaviciute, A., Jagielowicz, M., Fawcner-Corbett, D., Ashley, N., Cubitt, L., Mellado-Gomez, E., Attar, M., et al. (2018). Structural Remodeling of the Human Colonic Mesenchyme in Inflammatory Bowel Disease. *Cell* 175, 372–386.e17. <https://doi.org/10.1016/j.cell.2018.08.067>.
73. Tsukui, T., Sun, K.H., Wetter, J.B., Wilson-Kanamori, J.R., Hazelwood, L. A., Henderson, N.C., Adams, T.S., Schupp, J.C., Poli, S.D., Rosas, I.O.,

- et al. (2020). Collagen-producing lung cell atlas identifies multiple subsets with distinct localization and relevance to fibrosis. *Nat. Commun.* 11, 1920. <https://doi.org/10.1038/s41467-020-15647-5>.
74. Valenzi, E., Bulik, M., Tabib, T., Morse, C., Sembrat, J., Trejo Bittar, H., Rojas, M., and Lafyatis, R. (2019). Single-cell analysis reveals fibroblast heterogeneity and myofibroblasts in systemic sclerosis-associated interstitial lung disease. *Ann. Rheum. Dis.* 78, 1379–1387. <https://doi.org/10.1136/annrheumdis-2018-214865>.
75. Watson, B.R., Paul, B., Rahman, R.U., Amir-Zilberstein, L., Segerstolpe, A., Epstein, E.T., Murphy, S., Geistlinger, L., Lee, T., Shih, A., et al. (2025). Spatial transcriptomics of healthy and fibrotic human liver at single-cell resolution. *Nat. Commun.* 16, 319. <https://doi.org/10.1038/s41467-024-55325-4>.
76. Carmona-Rivera, C., Carlucci, P.M., Moore, E., Lingampalli, N., Uchtenhagen, H., James, E., Liu, Y., Bicker, K.L., Wahamoa, H., Hoffmann, V., et al. (2017). Synovial fibroblast-neutrophil interactions promote pathogenic adaptive immunity in rheumatoid arthritis. *Sci. Immunol.* 2, eaag3358. <https://doi.org/10.1126/sciimmunol.aag3358>.
77. Korsunsky, I., Wei, K., Pohin, M., Kim, E.Y., Barone, F., Major, T., Taylor, E., Ravindran, R., Kemble, S., Watts, G.F.M., et al. (2022). Cross-tissue, single-cell stromal atlas identifies shared pathological fibroblast phenotypes in four chronic inflammatory diseases. *Med.* 3, 481–518.e14. <https://doi.org/10.1016/j.medj.2022.05.002>.
78. Luo, H., Xia, X., Huang, L.B., An, H., Cao, M., Kim, G.D., Chen, H.N., Zhang, W.H., Shu, Y., Kong, X., et al. (2022). Pan-cancer single-cell analysis reveals the heterogeneity and plasticity of cancer-associated fibroblasts in the tumor microenvironment. *Nat. Commun.* 13, 6619. <https://doi.org/10.1038/s41467-022-34395-2>.
79. Salmon, H., Franciszkiewicz, K., Damotte, D., Dieu-Nosjean, M.C., Validire, P., Trautmann, A., Mami-Chouaib, F., and Donnadieu, E. (2012). Matrix architecture defines the preferential localization and migration of T cells into the stroma of human lung tumors. *J. Clin. Investig.* 122, 899–910. <https://doi.org/10.1172/JCI45817>.
80. Mariathasan, S., Turley, S.J., Nickles, D., Castiglioni, A., Yuen, K., Wang, Y., Kadel III, E.E., Koeppen, H., Astarita, J.L., Cubas, R., et al. (2018). TGFβ attenuates tumour response to PD-L1 blockade by contributing to exclusion of T cells. *Nature* 554, 544–548. <https://doi.org/10.1038/nature25501>.
81. Tauriello, D.V.F., Palomo-Ponce, S., Stork, D., Berenguer-Llergo, A., Badia-Ramentol, J., Iglesias, M., Sevillano, M., Ibiza, S., Cañellas, A., Hernandez-Momblona, X., et al. (2018). TGFβ drives immune evasion in genetically reconstituted colon cancer metastasis. *Nature* 554, 538–543. <https://doi.org/10.1038/nature25492>.
82. Cords, L., Engler, S., Haberecker, M., Rüschoff, J.H., Moch, H., de Souza, N., and Bodenmiller, B. (2024). Cancer-associated fibroblast phenotypes are associated with patient outcome in non-small cell lung cancer. *Cancer Cell* 42, 396–412.e5. <https://doi.org/10.1016/j.ccell.2023.12.021>.
83. Cheng, Y., Chen, X., Feng, L., Yang, Z., Xiao, L., Xiang, B., Wang, X., Liu, D., Lin, P., Shi, J., et al. (2025). Stromal architecture and fibroblast subpopulations with opposing effects on outcomes in hepatocellular carcinoma. *Cell Discov.* 11, 1. <https://doi.org/10.1038/s41421-024-00747-z>.
84. Shi, Y., Gao, W., Lytle, N.K., Huang, P., Yuan, X., Dann, A.M., Ridinger-Saison, M., DelGiorno, K.E., Antal, C.E., Liang, G., et al. (2019). Targeting LIF-mediated paracrine interaction for pancreatic cancer therapy and monitoring. *Nature* 569, 131–135. <https://doi.org/10.1038/s41586-019-1130-6>.
85. Orange, S.T., Leslie, J., Ross, M., Mann, D.A., and Wackerhage, H. (2023). The exercise IL-6 enigma in cancer. *Trends Endocrinol. Metabol.* 34, 749–763. <https://doi.org/10.1016/j.tem.2023.08.001>.
86. Lendahl, U., Muhl, L., and Betsholtz, C. (2022). Identification, discrimination and heterogeneity of fibroblasts. *Nat. Commun.* 13, 3409. <https://doi.org/10.1038/s41467-022-30633-9>.
87. Kraman, M., Bambrough, P.J., Arnold, J.N., Roberts, E.W., Magiera, L., Jones, J.O., Gopinathan, A., Tuveson, D.A., and Fearon, D.T. (2010). Suppression of antitumor immunity by stromal cells expressing fibroblast activation protein-α. *Science* 330, 827–830. <https://doi.org/10.1126/science.1195300>.
88. De Palma, M., Biziato, D., and Petrova, T.V. (2017). Microenvironmental regulation of tumour angiogenesis. *Nat. Rev. Cancer* 17, 457–474. <https://doi.org/10.1038/nrc.2017.51>.
89. Bartoschek, M., Oskolkov, N., Bocci, M., Lötvrot, J., Larsson, C., Sommarin, M., Madsen, C.D., Lindgren, D., Pekar, G., Karlsson, G., et al. (2018). Spatially and functionally distinct subclasses of breast cancer-associated fibroblasts revealed by single cell RNA sequencing. *Nat. Commun.* 9, 5150. <https://doi.org/10.1038/s41467-018-07582-3>.
90. Pelon, F., Bourachot, B., Kieffer, Y., Magagna, I., Mermet-Meillon, F., Bonnet, I., Costa, A., Givel, A.M., Attieh, Y., Barbazan, J., et al. (2020). Cancer-associated fibroblast heterogeneity in axillary lymph nodes drives metastases in breast cancer through complementary mechanisms. *Nat. Commun.* 11, 404. <https://doi.org/10.1038/s41467-019-14134-w>.
91. Bagaev, A., Kotlov, N., Nomie, K., Svekolkina, V., Gafurov, A., Isaeva, O., Osokin, N., Kozlov, I., Frenkel, F., Gancharova, O., et al. (2021). Conserved pan-cancer microenvironment subtypes predict response to immunotherapy. *Cancer Cell* 39, 845–865.e7. <https://doi.org/10.1016/j.ccell.2021.04.014>.
92. Gaggioli, C., Hooper, S., Hidalgo-Carcedo, C., Grosse, R., Marshall, J.F., Harrington, K., and Sahai, E. (2007). Fibroblast-led collective invasion of carcinoma cells with differing roles for RhoGTPases in leading and following cells. *Nat. Cell Biol.* 9, 1392–1400. <https://doi.org/10.1038/ncb1658>.
93. Labernadie, A., Kato, T., Brugués, A., Serra-Picamal, X., Derzsi, S., Arwert, E., Weston, A., González-Tarragó, V., Elosegui-Artola, A., Albertazzi, L., et al. (2017). A mechanically active heterotypic E-cadherin/N-cadherin adhesion enables fibroblasts to drive cancer cell invasion. *Nat. Cell Biol.* 19, 224–237. <https://doi.org/10.1038/ncb3478>.
94. Erdogan, B., Ao, M., White, L.M., Means, A.L., Brewer, B.M., Yang, L., Washington, M.K., Shi, C., Franco, O.E., Weaver, A.M., et al. (2017). Cancer-associated fibroblasts promote directional cancer cell migration by aligning fibronectin. *J. Cell Biol.* 216, 3799–3816. <https://doi.org/10.1083/jcb.201704053>.
95. Wu, T., Xiong, S., Chen, M., Tam, B.T., Chen, W., Dong, K., Ma, Z., Wang, Z., and Ouyang, G. (2023). Matrix stiffening facilitates the collective invasion of breast cancer through the periostin-integrin mechanotransduction pathway. *Matrix Biol.* 121, 22–40. <https://doi.org/10.1016/j.matbio.2023.05.006>.
96. Gong, Z., Li, Q., Shi, J., Wei, J., Li, P., Chang, C.H., Shultz, L.D., and Ren, G. (2022). Lung fibroblasts facilitate pre-metastatic niche formation by remodeling the local immune microenvironment. *Immunity* 55, 1483–1500.e9. <https://doi.org/10.1016/j.immuni.2022.07.001>.
97. Cohen, N., Mundhe, D., Deasy, S.K., Adler, O., Ershaid, N., Shami, T., Levi-Galibov, O., Wassermann, R., Scherz-Shouval, R., and Erez, N. (2023). Breast Cancer-Secreted Factors Promote Lung Metastasis by Signaling Systemically to Induce a Fibrotic Premetastatic Niche. *Cancer Res.* 83, 3354–3367. <https://doi.org/10.1158/0008-5472.CAN-22-3707>.
98. Deng, Y., Guo, C., Liu, X., Li, X., Liu, J., Liu, W., Chen, J., Huang, Z., Zhang, Y., Bi, X., et al. (2025). Single-cell transcriptomic profiling reveals liver fibrosis in colorectal cancer liver metastasis. *Exp. Mol. Med.* 57, 2517–2532. <https://doi.org/10.1038/s12276-025-01573-3>.
99. Li, T., Li, T., Liang, Y., Yuan, Y., Liu, Y., Yao, Y., and Lei, X. (2025). Colorectal cancer cells-derived exosomal miR-188-3p promotes liver metastasis by creating a pre-metastatic niche via activation of hepatic stellate cells. *J. Transl. Med.* 23, 369. <https://doi.org/10.1186/s12967-025-06334-4>.
100. Hanley, C.J., Waise, S., Ellis, M.J., Lopez, M.A., Pun, W.Y., Taylor, J., Parker, R., Kimbley, L.M., Chee, S.J., Shaw, E.C., et al. (2023). Single-cell analysis reveals prognostic fibroblast subpopulations linked to molecular and immunological subtypes of lung cancer. *Nat. Commun.* 14, 387. <https://doi.org/10.1038/s41467-023-35832-6>.
101. Chen, C., Guo, Q., Liu, Y., Hou, Q., Liao, M., Guo, Y., Zang, Y., Wang, F., Liu, H., Luan, X., et al. (2023). Single-cell and spatial transcriptomics reveal POSTN(+) cancer-associated fibroblasts correlated with immune suppression and tumour progression in non-small cell lung cancer. *Clin. Transl. Med.* 13, e1515. <https://doi.org/10.1002/ctm2.1515>.
102. Tong, W., Wang, T., Bai, Y., Yang, X., Han, P., Zhu, L., Zhang, Y., and Shen, Z. (2024). Spatial transcriptomics reveals tumor-derived SPP1

- p>induces fibroblast chemotaxis and activation in the hepatocellular carcinoma microenvironment.
- J. Transl. Med.*
- 22, 840.
- <https://doi.org/10.1186/s12967-024-05613-w>
- .
103. Jing, S.Y., Liu, D., Feng, N., Dong, H., Wang, H.Q., Yan, X., Chen, X.F., Qu, M.C., Lin, P., Yi, B., et al. (2024). Spatial multiomics reveals a sub-population of fibroblasts associated with cancer stemness in human hepatocellular carcinoma. *Genome Med.* 16, 98. <https://doi.org/10.1186/s13073-024-01367-8>.
 104. Pantaleo, J., Sjölund, J., Bolivar, P., Bocci, M., Phung, B., Malmberg, M., Jönsson, G.B., and Pietras, K. (2026). Spatial and temporal dynamics of cancer-associated fibroblast niches in breast cancer. *Breast Cancer Res.* 28, 21. <https://doi.org/10.1186/s13058-025-02183-7>.
 105. Qin, P., Chen, H., Wang, Y., Huang, L., Huang, K., Xiao, G., Han, C., Hu, J., Lin, D., Wan, X., et al. (2023). Cancer-associated fibroblasts undergoing neoadjuvant chemotherapy suppress rectal cancer revealed by single-cell and spatial transcriptomics. *Cell Rep. Med.* 4, 101231. <https://doi.org/10.1016/j.xcrm.2023.101231>.
 106. Huang, L., Lu, W., Wu, R., Li, Y., Ou, Z., Chen, J., Liu, Y., Yang, W., Xue, W., Mu, P., et al. (2025). Interferon-driven CAF reprogramming augments immunogenic response to neoadjuvant radiotherapy in colorectal cancer. *Cell Rep. Med.* 6, 102251. <https://doi.org/10.1016/j.xcrm.2025.102251>.
 107. Shiau, C., Cao, J., Gong, D., Gregory, M.T., Caldwell, N.J., Yin, X., Cho, J. W., Wang, P.L., Su, J., Wang, S., et al. (2024). Spatially resolved analysis of pancreatic cancer identifies therapy-associated remodeling of the tumor microenvironment. *Nat. Genet.* 56, 2466–2478. <https://doi.org/10.1038/s41588-024-01890-9>.
 108. Obradovic, A., Graves, D., Korner, M., Wang, Y., Roy, S., Naveed, A., Xu, Y., Luginbuhl, A., Curry, J., Gibson, M., et al. (2022). Immunostimulatory Cancer-Associated Fibroblast Subpopulations Can Predict Immunotherapy Response in Head and Neck Cancer. *Clin. Cancer Res.* 28, 2094–2109. <https://doi.org/10.1158/1078-0432.CCR-21-3570>.
 109. Jiang, J., Xu, C., Han, D., Lu, Y., Yang, F., Wang, J., Yan, X., Mu, X., Zhang, J., Jia, C., et al. (2024). Functional heterogeneity of cancer-associated fibroblasts with distinct neoadjuvant immunotherapy plus chemotherapy response in esophageal squamous cell carcinoma. *Biomark. Res.* 12, 113. <https://doi.org/10.1186/s40364-024-00656-z>.
 110. Ma, Z., Wang, Y., Wang, W., Sun, W., Wang, R., Ding, X., Chen, Z., Li, X., Li, Z., Ye, Y., et al. (2026). Lipid oxidation reprogramming in cancer-associated fibroblasts enhances CD8(+) T cell cytotoxicity and therapeutic response. *Cancer Cell* 44, 743–759.e8. <https://doi.org/10.1016/j.ccell.2026.01.012>.
 111. Denton, A.E., Roberts, E.W., Linterman, M.A., and Fearon, D.T. (2014). Fibroblastic reticular cells of the lymph node are required for retention of resting but not activated CD8+ T cells. *Proc. Natl. Acad. Sci. USA* 111, 12139–12144. <https://doi.org/10.1073/pnas.1412910111>.
 112. Roberts, E.W., Deonaraine, A., Jones, J.O., Denton, A.E., Feig, C., Lyons, S.K., Espeli, M., Kraman, M., McKenna, B., Wells, R.J.B., et al. (2013). Depletion of stromal cells expressing fibroblast activation protein- α from skeletal muscle and bone marrow results in cachexia and anemia. *J. Exp. Med.* 210, 1137–1151. <https://doi.org/10.1084/jem.20122344>.
 113. Hofheinz, R.D., al-Batran, S.E., Hartmann, F., Hartung, G., Jäger, D., Renner, C., Tanswell, P., Kunz, U., Amelsberg, A., Kuthan, H., and Stehle, G. (2003). Stromal antigen targeting by a humanised monoclonal antibody: an early phase II trial of sibrizumab in patients with metastatic colorectal cancer. *Onkologie* 26, 44–48. <https://doi.org/10.1159/000069863>.
 114. Scott, A.M., Wiseman, G., Welt, S., Adjei, A., Lee, F.T., Hopkins, W., Divgi, C.R., Hanson, L.H., Mitchell, P., Gansen, D.N., et al. (2003). A Phase I dose-escalation study of sibrizumab in patients with advanced or metastatic fibroblast activation protein-positive cancer. *Clin. Cancer Res.* 9, 1639–1647.
 115. Fu, H., Huang, J., Zhao, L., Chen, Y., Xu, W., Cai, J., Yu, L., Pang, Y., Guo, W., Su, B., et al. (2025). 177Lu-LNC1004 Radioligand Therapy in Patients with End-stage Metastatic Cancers: A Single-Center, Single-Arm, Phase II Study. *Clin. Cancer Res.* 31, 1415–1426. <https://doi.org/10.1158/1078-0432.CCR-24-3918>.
 116. Demetri, G.D., Luke, J.J., Hollebecque, A., Powderly, J.D., Spira, A.I., Subbiah, V., Naumovski, L., Chen, C., Fang, H., Lai, D.W., et al. (2021). First-in-Human Phase I Study of ABBV-085, an Antibody-Drug Conjugate Targeting LRRC15, in Sarcomas and Other Advanced Solid Tumors. *Clin. Cancer Res.* 27, 3556–3566. <https://doi.org/10.1158/1078-0432.CCR-20-4513>.
 117. Sherman, M.H., Yu, R.T., Engle, D.D., Ding, N., Atkins, A.R., Tiriach, H., Collisson, E.A., Connor, F., Van Dyke, T., Kozlov, S., et al. (2014). Vitamin D receptor-mediated stromal reprogramming suppresses pancreatitis and enhances pancreatic cancer therapy. *Cell* 159, 80–93. <https://doi.org/10.1016/j.cell.2014.08.007>.
 118. Kocher, H.M., Basu, B., Froeling, F.E.M., Sarker, D., Slater, S., Carlin, D., deSouza, N.M., De Paepe, K.N., Goulart, M.R., Hughes, C., et al. (2020). Phase I clinical trial repurposing all-trans retinoic acid as a stromal targeting agent for pancreatic cancer. *Nat. Commun.* 11, 4841. <https://doi.org/10.1038/s41467-020-18636-w>.
 119. Cho, B.C., Lee, J.S., Wu, Y.L., Cicin, I., Dols, M.C., Ahn, M.J., Cuppens, K., Veillon, R., Nadal, E., Dias, J.M., et al. (2023). Bintrafusp Alfa Versus Pembrolizumab in Patients With Treatment-Naive, Programmed Death-Ligand 1-High Advanced NSCLC: A Randomized, Open-Label, Phase 3 Trial. *J. Thorac. Oncol.* 18, 1731–1742. <https://doi.org/10.1016/j.jtho.2023.08.018>.
 120. Welsh, B.T., Faucette, R., Bilic, S., Martin, C.J., Schürpf, T., Chen, D., Nicholls, S., Lansita, J., and Kalra, A. (2021). Nonclinical Development of SRK-181: An Anti-Latent TGF β 1 Monoclonal Antibody for the Treatment of Locally Advanced or Metastatic Solid Tumors. *Int. J. Toxicol.* 40, 226–241. <https://doi.org/10.1177/1091581821998945>.
 121. Bockorny, B., Semenisty, V., Macarulla, T., Borazanci, E., Wolpin, B.M., Stemmer, S.M., Golan, T., Geva, R., Borad, M.J., Pedersen, K.S., et al. (2020). BL-8040, a CXCR4 antagonist, in combination with pembrolizumab and chemotherapy for pancreatic cancer: the COMBAT trial. *Nat. Med.* 26, 878–885. <https://doi.org/10.1038/s41591-020-0880-x>.
 122. Angevin, E., Tabernero, J., Elez, E., Cohen, S.J., Bahleda, R., van Laethem, J.L., Ottensmeier, C., Lopez-Martin, J.A., Clive, S., Joly, F., et al. (2014). A phase I/II, multiple-dose, dose-escalation study of siltuximab, an anti-interleukin-6 monoclonal antibody, in patients with advanced solid tumors. *Clin. Cancer Res.* 20, 2192–2204. <https://doi.org/10.1158/1078-0432.CCR-13-2200>.
 123. Chen, I.M., Donia, M., Chamberlain, C.A., Jensen, A.W.P., Draghi, A., Theille, S., Madsen, K., Hasselby, J.P., Toxværd, A., Høgdall, E., et al. (2023). Phase 2 study of ipilimumab, nivolumab, and tocilizumab combined with stereotactic body radiotherapy in patients with refractory pancreatic cancer (TRIPLE-R). *Eur. J. Cancer* 180, 125–133. <https://doi.org/10.1016/j.ejca.2022.11.035>.
 124. De Jesus-Acosta, A., Sugar, E.A., O'Dwyer, P.J., Ramanathan, R.K., Von Hoff, D.D., Rasheed, Z., Zheng, L., Begum, A., Anders, R., Maitra, A., et al. (2020). Phase 2 study of vismodegib, a hedgehog inhibitor, combined with gemcitabine and nab-paclitaxel in patients with untreated metastatic pancreatic adenocarcinoma. *Br. J. Cancer* 122, 498–505. <https://doi.org/10.1038/s41416-019-0683-3>.
 125. Catenacci, D.V.T., Junttila, M.R., Karrison, T., Bahary, N., Horiba, M.N., Nattam, S.R., Marsh, R., Wallace, J., Kozloff, M., Rajdev, L., et al. (2015). Randomized Phase Ib/II Study of Gemcitabine Plus Placebo or Vismodegib, a Hedgehog Pathway Inhibitor, in Patients With Metastatic Pancreatic Cancer. *J. Clin. Oncol.* 33, 4284–4292. <https://doi.org/10.1200/JCO.2015.62.8719>.
 126. Hingorani, S.R., Zheng, L., Bullock, A.J., Seery, T.E., Harris, W.P., Sigal, D.S., Braith, F., Ritch, P.S., Zalupski, M.M., Bahary, N., et al. (2018). HALO 202: Randomized Phase II Study of PEGPH20 Plus Nab-Paclitaxel/Gemcitabine Versus Nab-Paclitaxel/Gemcitabine in Patients With Untreated, Metastatic Pancreatic Ductal Adenocarcinoma. *J. Clin. Oncol.* 36, 359–366. <https://doi.org/10.1200/JCO.2017.74.9564>.
 127. Van Cutsem, E., Tempero, M.A., Sigal, D., Oh, D.Y., Fazio, N., Macarulla, T., Hitre, E., Hammel, P., Hendifar, A.E., Bates, S.E., et al. (2020). Randomized Phase III Trial of Pegvorhialuronidase Alfa With Nab-Paclitaxel Plus Gemcitabine for Patients With Hyaluronan-High Metastatic Pancreatic Adenocarcinoma. *J. Clin. Oncol.* 38, 3185–3194. <https://doi.org/10.1200/JCO.20.00590>.
 128. Bell, J.A., Davies, E.R., Brereton, C.J., Vukmirovic, M., Roberts, J.J.W., Lunn, K., Wickens, L., Conforti, F., Ridley, R.A., Ceccato, J., et al. (2024). Spatial transcriptomic validation of a biomimetic model of fibrosis

- enables re-evaluation of a therapeutic antibody targeting LOXL2. *Cell Rep. Med.* 5, 101695. <https://doi.org/10.1016/j.xcrm.2024.101695>.
129. Chakravarthy, A., Khan, L., Bensler, N.P., Bose, P., and De Carvalho, D. D. (2018). TGF-beta-associated extracellular matrix genes link cancer-associated fibroblasts to immune evasion and immunotherapy failure. *Nat. Commun.* 9, 4692. <https://doi.org/10.1038/s41467-018-06654-8>.
130. Li, J., Liu, H., Guo, Q., Zhang, Y., Li, J., Diao, T., Zheng, L., Deng, Z., Yang, Y., Chen, X., et al. (2026). Single-cell screens identify ADAM12 as a fibroblast checkpoint impeding anti-tumor immunity. *Cancer Cell* 44, 424–442.e14. <https://doi.org/10.1016/j.ccell.2025.12.018>.
131. Heide, J., Bilecz, A.J., Patnaik, S., Allega, M.F., Donle, L., Yang, K., Teich, E., Li, Y., Lin, Q., Kong, K., et al. (2025). NNMT inhibition in cancer-associated fibroblasts restores antitumour immunity. *Nature* 645, 1051–1059. <https://doi.org/10.1038/s41586-025-09303-5>.
132. Cheng, Y., Li, N., Zhu, B., Liu, X., Zhang, J., Xu, X., Hu, S., and Song, S. (2026). Diagnostic performance of [(18)F]F-FAPI-FUSCC-07 PET/CT in characterizing solitary pulmonary nodules: a head-to-head comparison. *Eur. J. Nucl. Med. Mol. Imag.* 53, 4549–4559. <https://doi.org/10.1007/s00259-026-07819-0>.
133. Wang, C., Xu, K., Li, W., Xu, L., Zhu, K., Dou, X., Liu, Y., Wang, Y., Zhang, X., and Yue, J. (2026). Assessment of liver fibrosis and tumor burden in hepatocellular carcinoma using [(18)F]AlF-NOTA-FAPI-04 PET/CT: a prospective study. *Eur. J. Nucl. Med. Mol. Imag.* 53, 3877–3890. <https://doi.org/10.1007/s00259-026-07804-7>.
134. Seban, R.D., Buvat, I., Champion, L., Bidard, F.C., Kieffer, Y., Vincent-Salomon, A., Peltier, A., and Mechta-Grigoriou, F. (2025). Beyond FAP: ANTXR1 as a novel target for PET imaging and radio-ligand therapy in immuno-oncology? *Eur. J. Nucl. Med. Mol. Imag.* 52, 1948–1950. <https://doi.org/10.1007/s00259-025-07126-0>.
135. Mascharak, S., Guo, J.L., Foster, D.S., Khan, A., Davitt, M.F., Nguyen, A. T., Burcham, A.R., Chinta, M.S., Guardino, N.J., Griffin, M., et al. (2023). Desmoplastic stromal signatures predict patient outcomes in pancreatic ductal adenocarcinoma. *Cell Rep. Med.* 4, 101248. <https://doi.org/10.1016/j.xcrm.2023.101248>.
136. Lin, Z., Li, G., Jiang, K., Li, Z., and Liu, T. (2024). Cancer therapy resistance mediated by cancer-associated fibroblast-derived extracellular vesicles: biological mechanisms to clinical significance and implications. *Mol. Cancer* 23, 191. <https://doi.org/10.1186/s12943-024-02106-8>.
137. Albrecht, L.J., Höwner, A., Griewank, K., Lueong, S.S., von Neuhoff, N., Horn, P.A., Sucker, A., Paschen, A., Livingstone, E., Ugurel, S., et al. (2022). Circulating cell-free messenger RNA enables non-invasive pan-tumour monitoring of melanoma therapy independent of the mutational genotype. *Clin. Transl. Med.* 12, e1090. <https://doi.org/10.1002/ctm2.1090>.
138. Moss, J., Magenheimer, J., Neiman, D., Zemmour, H., Loyfer, N., Korach, A., Samet, Y., Maoz, M., Druid, H., Arner, P., et al. (2018). Comprehensive human cell-type methylation atlas reveals origins of circulating cell-free DNA in health and disease. *Nat. Commun.* 9, 5068. <https://doi.org/10.1038/s41467-018-07466-6>.